Supplementary Information

This Supplementary Information accompanies the main manuscript and comprises eleven sections: S1 the checklist of fields and measurement structures that must be newly recorded, S2 the data dictionary, S3 the curation specification, S4 the seven evaluation splits, S5 the quantum-chemical chirality descriptor definitions, S6 the models and tuning protocol, S7 the full per-fold result tables, S8 the negative controls, S9 the reproducibility gate sequence, S10 the ten disclosed limitations, and S11 the additional analyses.

S1. Checklist of fields and data structures that must be newly measured

This section is the supplementary-material version of the data-collection checklist; the main text refers to it as Supplementary Information S1.

The conclusion in one sentence: the conclusion of this study is not "machine learning does not work on this problem", but **"the structure of the available public data is insufficient to support cross-laboratory prediction"**.

S1.1 The 7 missing condition fields (CMRT has only the `i-PrOH proportion` and the `flow rate`)

#	field	consequence of omitting it (measured)	collection cost
1	Buffer salt type and concentration	lumped into the unexplained 38.5 %	very low
2	pH (aqueous, or the apparent pH when additives are present)	same as above	very low
3	**Column temperature**	same as above; the mechanistic information in ΔΔH/ΔΔS is also lost	low (column-oven readout)
4	Gradient program / isocratic flag	the definition of k does not hold and back-solving t_0 fails (57 pairs with $k \le 0$ are already removed from this table)	low
5	Additives (TFA / DEA / ammonium formate, etc.)	lumped into the unexplained variance	very low
6	**Measured dead time t_0**	currently estimated (φ = 0.62); the uncertainty of α at the low-retention end is amplified to **±37 %**	low (uracil/thiourea method)
7	Column batch / serial number	the "laboratory effect" cannot be decomposed into "batch vs method"	low

S1.2 The 3 missing measurement structures (more important than the fields)

1. **The same molecule × multiple conditions**: CMRT has 10,126 molecules / 10,525 rows ⇒ the molecular main effect `a_m` and the interaction `h` cannot be separated, while `h` accounts for **76.95 %**. Requirement: at least **one rectangular grid of 30–50 molecules × ≥4 conditions** (the 43 × 12 grid of DG25 is precisely what made it possible to quantify `h`).

2. **Replicate measurements**: each cell currently holds exactly 1 observation ⇒ interaction and noise are inseparable. Requirement: **≥3 replicates** for the same (molecule, column, mobile phase, temperature) (2 on the same day + 1 on another day).
3. **Record the "not resolved" outcomes**: only **0.16 %** of CMRT samples have `α ≡ 1`, whereas the DG25 test subset (**449 rows**: CMRT overlaps removed + structureless molecules removed) gives **49–72 % per column and 58 % overall** ⇒ selection bias in CMRT. Requirement: record failed attempts as well (`α = 1.00`, `Rs < 1.5`, peaks not resolved), and distinguish "tried but not resolved" from "not attempted".

S1.3 Collection template (can be handed to experimentalists as is)

```
Required (existing fields):
  analyte SMILES / name, CSP column name, flow rate, i-PrOH (or organic phase) proportion,
  retention times of the two enantiomers, elution order, source (DOI / this laboratory)

New required fields (S1.1):
  buffer salt type + concentration, pH, column temperature (°C), isocratic/gradient (give the
program), additives, measured t₀, column lot number

New structural requirements (S1.2):
  □ ≥3 replicates per cell (2 same-day + 1 on another day)
  □ build a "molecule × condition" grid wherever possible (≥30 molecules × ≥4 conditions)
  □ record failed separations too, and distinguish "tried but not resolved" from "not
attempted"
```

S2. Data dictionary

S2.1 Main modeling table `data/curated/cmrt_pairs.csv` (10,525 rows × 38 columns; one row = one exact enantiomer pair within one condition cell)

The released table has **38 columns**; column 38 is `flag_no_separation` (the 17 rows whose α is exactly 1).

#	column	dtype	definition / unit	note (measured)
1	`pair_id`	str	row ID in the main table (`P%05d`, 0–10524)	10,525 unique values
2	`conn`	str	InChIKey connectivity layer (first 14 characters) = stereo-stripped molecular identity	10,126 unique values
3	`murcko`	str	Murcko scaffold (`Chem.MurckoDecomp ose`)	2,769 unique values; **an empty string for acyclic molecules is normal** (0 empty strings in this table)
4	`smi_S`	str	canonical isomeric SMILES of the CIP-**S** enantiomer (RDKit)	10,129 unique values
5	`smi_R`	str	canonical isomeric SMILES of the CIP-**R** enantiomer	10,129 unique values
6	`cip_S`	str	CIP signature on the S side (ordered by canonical atom rank)	`S` 10,516; the other 9 are `SS`/`SR`/`SRR`/`RS` (2–3 stereocenters)
7	`cip_R`	str	CIP signature on the R side	`R` 10,516; the rest are the corresponding

#	column	dtype	definition / unit	note (measured)
				complementary signatures
8	`n_stereo`	int	number of stereocenters	1: 10,516; 2: 8; 3: 1
9	`Column`	str	column name (Daicel trade name)	25 unique values
10	`surface`	str	polysaccharide backbone: `Amylose` / `Cellulose`	2 unique values
11	`subs_code`	str	substituent code a–g (verified against the official Daicel pages, see S2.3)	7 unique values
12	`particle_um`	float	particle size (μm)	4 unique values: 1.6 / 3 / 5 / 10
13	`family`	str	backbone × substituent family name (e.g. `Cellulose-a`)	10 unique values
14	`ipr`	float	i-PrOH volume fraction (dimensionless, 0–1)	measured 0.0–0.5; 77 distinct values
15	`speed`	float	flow rate (mL/min)	measured 0.1–13.0; 30 distinct values
16	`t0`	float	**estimated** dead time (min), t_o = `V_m/F`, `V_m = 0.62·V_c`, `V_c = 4.155 cm³`	measured 0.198–25.76; 30 distinct values (tracking `speed`)
17	`k_S`	float	retention factor of the CIP-S enantiomer, $k = (RT - t_o)/t_o$	dimensionless
18	`k_R`	float	retention factor of the CIP-R enantiomer	dimensionless
19	`alpha_true`	float	α = `k_S / k_R` (chromatographic definition, **not** RT_2/RT_1)	measured 0.0311–24.51
20	`y`	float	**main target variable** `y = ln α_true = ln[k(S)/k(R)]` (directed, CIP-directed)	SD 0.4121; median +0.0455; range −3.471 to +3.199
21	`EMO`	int	elution-order target `EMO = 1{y > 0}` (CIP-S elutes later)	mean 0.5067 (no systematic bias)
22	`rt_S`	float	retention time on the S side (min)	raw observation
23	`rt_R`	float	retention time on the R side (min)	raw observation
24	`dRT_over_t0`	float	$(RT_S - RT_R)/t_o$ normalized retention difference (**directed**)	sign agreement with `y` **99.97 %** (an earlier description says "100 %"; measured 99.97 %)
25	`rt_ratio_spike`	float	undirected RT ratio `max/min` (legacy audit-period quantity, **not used in modeling**)	8533 distinct values
26	`n_rep`	int	number of records aggregated into this condition cell	2: 10,050; 4: 444; 6: 20; 8: 9; 3: 2 (**mostly even**)

#	column	dtype	definition / unit	note (measured)
27	`kS_n`	int	effective number of replicates for k on the S side	1: 10,050; 2: 446; 3: 20; 4: 9
28	`kR_n`	int	effective number of replicates for k on the R side	1: 10,052; 2: 444; 3: 20; 4: 9
29	`kS_rsd`	float	relative standard deviation of k on the S side	**96.2 % are exactly 0** (undefined when `kS_n=1`); the non-zero ones have n = 396, median 0.0546, maximum 0.947
30	`kR_rsd`	float	relative standard deviation of k on the R side	**96.4 % are exactly 0**; otherwise as above
31	`doi`	str	repaired source-publication DOI	530 unique values
32	`doi_raw`	str	original DOI (**by design** the pre-repair value)	⚠ **measured to be identical to `doi` row by row** (10,525/10,525) ⇒ the original strings survive only in `cmrt_measurements.csv` (see the two caveats under step 3 in S3)
33	`cond_matched`	bool	whether the two enantiomers were measured within the **same condition cell**	**True** for the whole table (the final main table keeps only same-cell pairings)
34	`flag_alpha_extreme`	bool	`abs(ln α) > ln 20` (suspected extraction error)	**3** rows
35	`flag_cond_pooled`	bool	different condition cells, pairs pooled within a column (impure α conditions)	**False** for the whole table (= no such rows)
36	`flag_dup_pair`	bool	the same `(molecule, column)` appears under several conditions	**308** rows
37	`flag_cip_ambiguous`	int	CIP signatures not complementary, so the S/R sign semantics are unreliable	**1** row (`P04270`, column `IAH`, `n_stereo = 2`; 0.01 %)
38	`flag_no_separation`	bool	α equal to 1 within `rtol=1e-9` ⇒ EMO undefined	**17** rows (0.16 %; 15 of them have α exactly 1.0, 2 differ by 2 ulp)

Derived statistics of the main table (directly citable): 10,126 molecules; median **1** row per molecule (mean 1.039, maximum 8); median **18** rows per source publication (1–120); median **247** pairs per column (13–2216; median 245 molecules per column).

S2.2 Record-level table `data/curated/cmrt_measurements.csv` (25,866 rows × 21 columns)

column	definition
`SMILES`, `smi_iso`, `smi_flat`	raw / canonical isomeric / stereo-stripped SMILES
`conn`, `inchikey`, `murcko`	connectivity layer / full InChIKey / scaffold
`cip`, `n_stereo`, `n_stereo_undef`	CIP signature / number of stereocenters / number of undefined stereocenters
`RT`, `Speed`, `i-PrOH_proportion`, `Column`	raw observed fields
`t0`, `k`	estimated dead time, retention factor
`doi`, `doi_raw`	repaired / original DOI (**this table is the authoritative home of `doi_raw`**)
`flag_rt_gt_60`	RT > 60 min: **576** rows (flagged, not deleted)
`flag_speed_out`	Speed > 4 mL/min: **6** rows (including 13 mL/min; flagged for review)
`flag_k_nonpos`	`k ≤ 0`: **108** rows (the main table removes 57 pairs on this basis)
`flag_undef_stereo`	undefined stereochemistry: **370** rows

S2.3 Column metadata `data/curated/columns.csv` (25 rows × 7 columns)

Columns: `Column, surface, particle_um, subs_code, family, subs_meaning, n_pairs`.

Chemical identity of the substituents (verified one by one against the official Daicel pages):

code	substituent	Amylose side	Cellulose side
a	3,5-dimethylphenylcarbamate	AD / AD3 / ADH / IA / IAH / IAU	IB / OD / OD3 / ODH
b	3,5-dichlorophenylcarbamate	IE	IC / IC3 / ICH
c	4-methylbenzoate	—	OJ / OJ3 / OJH
d	(S)-1-phenylethylcarbamate (**the substituent itself carries a stereocenter**)	AS / ASH	—
e	3-chlorophenylcarbamate	ID (single column)	—
f	3-chloro-4-methylphenylcarbamate	AZ / IF / IF3	OZ3
g	3-chloro-5-methylphenylcarbamate	IG (single column)	—

Naming pattern: the prefix `A*` = amylose / `O*` = cellulose; the suffix `3` → 3 μm, `H` → 5 μm, `U` → 1.6 μm; bare names are the coated type = 10 μm, and the bonded type `IA–IG` = 5 μm. **The matched pairs usable for same-substituent, different-backbone extrapolation are still 3 groups**: {Amylose-a, Cellulose-a}, {Amylose-b, Cellulose-b}, {Amylose-f, Cellulose-f}, covering **81.0 %** of the main table.

S2.4 Split indices `data/curated/splits/*.csv` (8 files)

The main text defines **7 protocols** (S1 / S2 / S2b / S2c / S3 / S3′ / S4); **8 files** are on disk, the extra `S4b_ecfp_class.csv` being an alternative version ("5 folds by ECFP4 clustering") that **the main text does not cite**. Per-fold sizes are given in S4.

S2.5 Descriptor tables (three sets)

file	size	content
`data/curated/l1a_descriptors.csv`	**9,968 rows × 111 columns** (48×2 descriptors + 15 diagnostic columns)	L1a-FF force-field-level pseudoscalars
`data/curated/l1cf_descriptors.csv`	**9,968 rows × 96 columns** (89 `cf_*` descriptors + 7 metadata columns)	L1-CF local chiral-frame shape
`data/curated/cmcds_l1b.csv`	**10,887 rows × 76 columns**	L1b: TD-DFT ECD data from CMCDS (20 states E/λ/R + summary)
`data/curated/l1a_failures.csv` / `l1cf_failures.csv`	**149 rows** each	molecules for which no descriptor could be built (`conn, err`)
`data/curated/l1a_per_column.csv`	48 rows	per-column degeneracy rate and sign diagnostics
`data/curated/csp_descriptors.csv`	25 rows	self-built CSP descriptors (167 dimensions)
`data/curated/zeroorder_scan.csv`	196 rows	per-column zero-order correlations for the three descriptor generations (`family, col, r, r_emo, signed`)

S2.6 Other artifacts on disk (every number cited in the paper comes from these)

`curation_stats.json` (machine-readable statistics); `s2c_emo.csv`, `s3_emo.csv` (EMO gradient); `leave_one_doi_summary.csv` (main S2c results); `failure_decomposition.csv` (86-row per-fold decomposition); `learning_curve_papers.csv`, `within_paper_split.csv` (same vs. different laboratory); `reproducibility_components.csv`, `paper_offset_predictability.csv`, `offset_power_curve.csv` (ceiling and power curve); `condition_pairwise.csv` (model-free condition pairs); `interaction_ceiling.csv` (additive upper bound); `external_odrh_result.csv`, `external_all_columns.csv` (external validation); `t0_sensitivity.csv`, `recurate_phi.csv` (dead-time sensitivity); `l1a_signal_ablation.csv`, `l1cf_signal_ablation.csv`, `l1b_signal_ablation.csv` (three-generation ablation).

S3. Curation specification: audit-ready detail of the 9 steps

The **single authoritative implementation** of the curation specification is `scripts/01_curate_cmrt.py`; the gate is `scripts/_qa_01_curate.py` (**all 67 checks pass**). Arrows denote "input → output".

#	step	input → output	rule / basis	consequence (if not done)
1	Drop nulls	25,867 → 25,867	the five columns `SMILES / RT / Speed / i-PrOH / Column` have **zero missing values**	none
2	Remove `RT ≤ 0`	25,867 → **25,866**	a negative retention time is physically meaningless (**1 record**)	a negative RT would contaminate `k`, t_0 and α
3	**DOI repair**	**1,307 records / 24 unique DOIs** rewritten	about 5 % of the records write `-` as `/` in the DOI suffix (`10.1038/s41467/019/13705/1` →	without this, "100 % of DOIs are traceable" is an empty claim

#	step	input → output	rule / basis	consequence (if not done)
			`10.1038/s41467-019-13705-1`). **Rule**: act only when the suffix contains `/`; `re.match(r"^(10\.\d{4,9})/(.+)$", d)`, replacing every `/` after the first one with `-`; assert that **the maximum number of slashes in `doi` = 1**	
4	RDKit canonicalization	23,793 unique raw SMILES → **23,781** unique isomeric SMILES; **0 parse failures**	`MolFromSmiles` → `MolToSmiles(isomericSmiles=True)`	one molecule is split across several rows because of drawing differences
5	**Estimate t_o, compute `k`**	per record, `t_o = V_m / Speed`	the dataset has **no measured dead time**. `Column_info.xlsx` states an internal diameter of 4.6 mm / column length of 250 mm for all 25 columns ⇒ `V_c = 4.155 cm³`; with a porosity basis of **φ = 0.62** ⇒ `V_m = 2.576 cm³`; the sensitivity scan is in Section 2.8 / S10 item 7	without estimating t_o, neither `k` nor α can be defined
6	**Condition-cell folding**	25,866 records → **24,624 condition cells**	a condition cell = `(molecule, column, i-PrOH, flow rate)`; **replicates must first be aggregated within the cell, and pairing must then be done within the cell**	otherwise α is a mixture of two different mobile phases
7	**Exact-enantiomer determination**	11,179 candidates → **10,582 exact enantiomer pairs**	**two-method cross-validation**: M1 stereo inversion (`invert_stereo` + canonical-SMILES comparison), M2 the InChI `/t` layer (same `/t` with a different `/m`, or `/t` fully inverted character by character `+/-`). **The two methods disagree in 0 cases** and jointly remove **597 (5.3 %) diastereomers**	pairing on the InChIKey connectivity layer alone would treat diastereomers as enantiomers and directly contaminate the target variable (their ΔΔG bears no antisymmetric relation to the pseudoscalar)
8	Remove `k ≤ 0`	**57 pairs** discarded (108 rows at record level, `flag_k_nonpos`)	`RT < t_o` is physically impossible ⇒ extraction error or gradient elution	an undefined `k` propagates into an undefined α
9	**CIP orientation**	10,582 → **10,525 pairs** in the main table	R/S is assigned by the new RDKit CIP (`FindMolChiralCenters(useLegacyImplementation=False)`); **not by elution order**;	an undirected α would discard the elution order and would not align with a pseudoscalar that changes sign strictly

#	step	input → output	rule / basis	consequence (if not done)
			the target variable is `y = ln[k(S)/k(R)]`	

S3.1 What the 597 removed pairs look like

About **98 %** are cases where "only 1 of 2 stereocenters is inverted" (apparently epimers of each other); the rest involve 3 or 4 stereocenters. The determination was validated on 7 control groups, covering a single stereocenter, a single stereocenter + E alkene, cis/trans rings, full inversion of both stereocenters, inversion of only one of two stereocenters, and a **meso compound** (cis-1,4-cyclohexanediol is its own mirror image, correctly treated as a "diastereomer").

S3.2 Two definition caveats that readers can check themselves (identified in the self-audit for this study)

1. **`doi_raw` is not faithful in the main table**: `doi_raw` in `cmrt_pairs.csv` is identical to `doi` row by row (10,525/10,525); the original (corrupted) DOIs exist only in `cmrt_measurements.csv`. **To reproduce the DOI-repair statistics, use the record-level table.**
2. **1 record has the literal DOI `"s"`** (column `ADH`, `pair_id = P00570`): a defect in the source data; the repair rule (which requires a `/` in the suffix) does not apply to it, so it **cannot be repaired**. It accounts for 0.01 % of the main table, and its `doi_raw` is likewise `"s"`. **It is recommended to state this in the data-availability declaration alongside the 597 removed pairs.**
3. The assertion in step 3 is "**maximum** number of slashes = 1" (`assert r["doi"].str.count("/").max() == 1`), **not** "exactly 1 slash per row"; a value with 0 slashes such as `"s"` is therefore not caught.

S4. The seven splits: definitions and per-fold sizes

Terminology: in this dataset **one source publication = one DOI = one study**, so `leave-one-paper-out` and `leave-one-study-out` are synonymous and the former is used throughout. Every split carries a `dup_group = (molecule, column)` key so that the same measurement cannot appear on both sides.

protocol	grouping variable	folds	file on disk	per-fold size (min / med / max rows)	purpose
S1 random	none (point-wise 80/20)	2	`S1_random.csv`	2105 / 5262.5 / 8420	reproduces common practice in the field; **deliberately leaky** (53 `dup_group` values measured to repeat across folds)
S2 grouped	`(conn, Column)`	5	`S2_group_dup.csv`	2105 / 2105 / 2105	removes same-molecule same-column leakage
S2b leave-a-	`doi`, 10 groups	10	`S2b_leave_paper.csv`	1052 / 1052.5 /	⚠ **not leave-**

protocol	grouping variable	folds	file on disk	per-fold size (min / med / max rows)	purpose
batch-of-papers				1053	**one-paper-out**: ~53 papers per fold, so the paper effect is averaged away within a fold (the fold means of y differ by only ±0.047, whereas the between-paper offset SD is 0.256) ⇒ **systematically optimistic**
S2c true leave-one-paper-out	`doi`, **one paper per fold**	**530**	`S2c_leave_doi.csv`	1 / 18 / 120	**the primary reported protocol** (`η²(doi) = 0.3854`)
S3 leave-one-column	`Column`	25	`S3_leave_column.csv`	13 / 247 / 2216	cross-CSP extrapolation
S3′ factor extrapolation	substituent code × backbone	6 tasks	`S3p_factor.csv` (**long format, 17,046 rows, no `fold` column**)	see the table below	holds out "selector combinations absent from the training set"
S4 leave-scaffold	`murcko` / ECFP4 clustering	5	`S4_murcko.csv`	2105 / 2105 / 2105	chemical-class extrapolation
(extra) `S4b`	ECFP4 clustering	5	`S4b_ecfp_class.csv`	2105 / 2105 / 2105	alternative version **not cited in the main text**

The 6 tasks of S3′ (train → test rows):

task	train	test
`S3p_a_amylose→cellulose`	3492	2738
`S3p_a_cellulose→amylose`	2738	3492
`S3p_b_amylose→cellulose`	388	1367
`S3p_b_cellulose→amylose`	1367	388
`S3p_f_amylose→cellulose`	434	104
`S3p_f_cellulose→amylose`	104	434

S4.1 The three fold-count definitions

Provenance and status: the numbers in this section come from an independent recomputation and were confirmed by a local recomputation (same result, see below). ✅ **The verdict is in**: all three numbers **are true and refer to three different samples**; both definitions should be stated (primary fold count = **530**, with the 523 and 428 subsets explained separately).

All three numbers are correct, but they refer to three different samples (local recomputation, commands and results):

#	basis	folds	rows	papers	definition	source
①	**split level** (authoritative)	**530**	10,525	530	one DOI = one fold (1–120 rows, median 18)	`data/curated/splits/S2c_leave_doi.csv`

#	basis	folds	rows	papers	definition	source
②	**actual evaluation subset of the descriptor regression arm**	**523**	**10,363**	523	requires molecules to lie within the L1-CF ∩ L1a-FF coverage (9,968 molecules)	the log line of `scripts/15` = where the 523 value comes from
③	**folds entering the per-fold statistics + global R^2 summary**	**428**	**10,193**	428	additionally requires ≥ **5** rows within a fold (`--min-n` default 5) ⇒ 95 folds / 170 rows discarded	`data/curated/leave_one_doi_summary.csv`

Recomputation checks: `523 - 95 = 428`, `10,363 - 170 = 10,193` ✓; `η²(fold)` = **0.388746** (the 10,363-row subset, i.e. 0.3887), **0.381109** (the 10,193-row subset), **0.385398** (the full 10,525-row table / 530 folds) ✓.

🔴 **Two facts that belong together** (otherwise the discrepancy is obvious at a glance): 1. **EMO 52.85 % and R^2 −0.107 come from two different row samples**: the EMO arm (`scripts/18`) uses **all 10,525 rows / 530 folds**; the R^2 arm (`scripts/15`) has **10,363 rows / 523 papers** after descriptor-coverage filtering, while the `n=10,193 / n_fold=428` summarized in `leave_one_doi_summary.csv` is the result of a further "≥5 rows within a fold" filter. **Writing 523 as the "fold count" in Table 2 is inaccurate** (523 is the number of papers in that arm, not the number of folds in the split). 2. **The minimum fold size in `fold_stats` is 5, not 20**: `--min-n` defaults to **5** in `scripts/15` (the function default of 20 in `scripts/13.fold_stats` takes effect only when that argument is not passed). Moreover, the `--min-n` help text of `scripts/15` says "does not affect the global R^2", but the global R^2 of `scripts/13.summarize()` **is reconstructed by weighting the filtered folds** (numerator covering 10,193 rows, denominator `var(y)` taken over 10,363 rows) ⇒ **the help text does not match the implementation**. 3. ⇒ **Verdict**: the main text states **both definitions** — **the primary fold count is 530** (= the true number of folds in the split files, also consistent with the 10,525 rows of the EMO arm), with a note in Methods / the table caption that "the descriptor arm is filtered by coverage to **523 papers / 10,363 rows**, and the per-fold statistics and summary R^2 further require ≥5 rows within a fold ⇒ **428 folds / 10,193 rows**". 523 **must not** be presented on its own as the "fold count". 4. **A small definition defect**: the global R^2 of `scripts/13.summarize()` **accumulates the numerator over only 428 folds / 10,193 rows, yet uses the whole 10,363-row block for the denominator `var(y)`** (0.169995 vs 0.169664, a difference of **0.2 %**) ⇒ strictly, the global R^2 should be computed on the same set; the magnitude is negligible.

S4.2 Why "leave-one-column-out" cannot serve as the main evidence for cross-column transfer

At the molecular level CMRT is almost 25 non-overlapping single-column subsets (only **0.3 %** of molecules cross columns), so leave-one-column-out **cannot distinguish** "the CSP descriptors transfer" from "the model merely generalizes to new molecules"; factor extrapolation (S3′) is free of this confound, so the main text uses S3′ as the evidence for transfer across stationary phases.

S5. Three generations of quantum-chemical chirality descriptors: complete definitions

S5.1 `L1a-FF` — stereocenter-anchored force-field-level pseudoscalars (`scripts/06_chirality_descriptors.py`)

Geometry: ETKDG multiple conformers + MMFF94 optimization, fixed seed **`SEED = 0xC1A1`**; Boltzmann weighting (`RT = 1.98720425864083e-3 × 298.15` kcal/mol); mass weights use atomic masses. Conformer-count threshold **`MIN_CONF = 1`**.

Construction (the four directly bonded neighbours with the highest CIP rank, `n0 > n1 > n2 > n3`, serve as the branches):

```
v_j = r_{n_j} − r_k                        (k = stereocenter, j = 0..3)
vol_w    = det[ v1−v0, v2−v0, v3−v0 ]      ← signed volume (pseudoscalar)
voln_w   = as above but the three vectors are normalized first  ← purely angular chirality
tp{ijl}_w = u_i · (u_j × u_l)              (all triples {i,j,l} ⊂ {0,1,2,3})
```

8 atomic scalar weights `w ∈ {mass, gast, absq, deg, ring, arom, en, cnt}` (mass / Gasteiger charge / |charge| / degree / ring / aromatic / Pauling electronegativity / count); two aggregation modes, **`_bw` (with Boltzmann weights) and `_sc` (single conformer)**.

family	dimensions	definition	note
`vol_*`	8	`det[u₁−u₀, u₂−u₀, u₃−u₀]`	⚠ **strictly linearly redundant** with `tp***`
`voln_*`	8	as above but the three vectors are normalized first (purely angular chirality)	independent
`tp{012,013,023,123}_*`	32	branch triple products, three of the four	independent; `tp012` and `vol` have **opposite signs** (permutation parity)
Total	**48**	(plus 48 `_sc` single-conformer columns)	true rank ≤ **40**

Degeneracy and redundancy that must be disclosed (per-column measurements in S7-J):

- **12/48 columns (25 %) are structurally always zero**: all `*_ring` and `*_arom` weight columns (no ring/aromatic atom among the branches ⇒ all weights zero); of these, **6 columns** are zero for ≥90 % of molecules (`zero_frac`: `tp013_arom` 0.962, `tp023_arom` 0.948, `tp123_arom` 0.946, `tp013_ring` 0.916, `tp123_ring` 0.915, `tp023_ring` 0.911). Downstream modeling always drops degenerate columns.
- **8 columns are strictly linearly redundant**: `vol_w = tp123_w − tp023_w + tp013_w − tp012_w` (for the same weight) ⇒ the 48 columns have **true rank ≤ 40**; `--no-vol` produces the clean version.
- ⚠ **The `np.sign(0) = 0` trap**: always-zero columns were once counted as "sign-inconsistent", which pushed the "all-same-sign fraction" down from 75.8 % to 62 %; `06` now reports `n_zero_col / n_live / cons_med_nz` separately.
- **The sign factor is almost constant**: the pseudoscalar `≈ s × even quantity`, and `cip_S` is `S` for 10,516/10,525 pairs (single stereocenter in 99.9 % of them) ⇒ `s` has almost no variation ⇒ **the information carried by the pseudoscalar is essentially the same as that of its even-quantity version**. This is the mechanistic evidence that "the chirality columns carry no extra information".

- **Coverage**: **10,117** molecules with a single stereocenter ⇒ **9,968 successes (98.5 %)**; **149 failures** (1.5 %), **all of them sulfur stereocenters (sulfoxides, Z = 16)** — a deliberate exclusion that must be declared (see `l1a_failures.csv` for the failed cases).
- **Naming**: this script implements the **force-field-level (FF)** precursor, denoted **`L1a-FF`** throughout this paper.

S5.2 `L1-CF` — whole-molecule shape in a local chiral frame (`scripts/06c_chiral_frame_descriptors.py`)

Motivation: the L1a-FF of `06` has only "4 branch vectors around the stereocenter × 8 weights" and **never touches the rest of the molecule**. Chiral recognition by a CSP happens in the molecule–stationary-phase contact layer (about 3–7 Å), so what is needed is "**the chiral arrangement of the rest of the molecule relative to the stereocenter**", not "the orientation of each of the four branches".

Reference-frame construction (taking the four CIP-rank-ordered neighbour atoms `n0 > n1 > n2 > n3` of stereocenter `k`):

```
v_j = r_{n_j} − r_k
e1 = unit(v0)
e2 = unit(v1 − (v1·e1) e1)        # orthogonalization
e3 = e1 × e2                      # pseudovector: under mirror reflection e3 → −e3
s  = sign(det[v0, v1, v2])        # chirality sign (isomorphic to R/S; nearly constant over
the whole table)
```

For **every heavy atom i** other than k: `d_i = r_i − r_k`; `z_i = d_i · e3` (pseudoscalar, changes sign under mirror reflection); `η_i = s · z_i` (**even quantity**); `ρ_i = √(|d_i|² − z_i²)` (in-plane radius); `r_i = |d_i|` (radial distance).

The 89 `cf_*` descriptors (column count measured from the released CSV):

block	content	dimensions
A Atom-group moments	7 chemical groups (`all` / `C` / `het` / `arom` / `ring` / `hb` / `halo`) × 8 functionals: `n` (count), `m1` (Ση), `m1n` (η/n), `m1abs` (Σabs(η)/n), `m2` (Ση²), `m3` (Ση³), `meanrho` (Σρ/n), `meanr` (Σr/n)	**56**
B Signed radial profile	7 radial shells (starting at 2.0/2.8/3.6/4.6/5.8/7.2/9.0 Å) × {mass, mass·η, abs(charge)·η} ⇒ `Σ_{r_i∈shell} m_i·η_i` (a **signed radial distribution function**)	**21**
C Global scalars	`cf_n_heavy, cf_n_het, cf_n_arom, cf_n_ring, cf_n_rotb, cf_rg, cf_rmax, cf_rmean, cf_rg_het, cf_rg_arom, cf_rg_rel`	**11**
Chirality sign	`cf_s` (≈ +1, nearly constant over the whole table)	**1**
Total		**89**

Coverage is the same as for L1a-FF (**9,968 / 10,117 = 98.5 %**, the same 149 sulfur stereocenters fail).

S5.3 `L1b` — TD-DFT ECD rotational strengths (`scripts/12_build_cmcds.py`, data from CMCDS)

Data source: CMCDS (`data/raw/cmcds/CMCDS_DATASET.csv`, 7.2 MB, MD5 `181c1015f2befeeedcf15da5d8494650`) ⇒ parsed into `data/curated/cmcds_l1b.csv` (**10,887 rows × 76 columns**).

Construction: **20 excited states** per molecule (`E1..20` excitation energy in eV, `lam1..20` wavelength in nm, `R1..20` rotational strength `R(velocity)`), condensed into summary pseudoscalars:

column	meaning	median
`R_sum`	Σ R (net rotational strength, **pseudoscalar**)	−0.1031
`Rw_lam`	rotational-strength centroid wavelength (pseudoscalar)	−0.1595
`R_absmean` / `R_absmax`	mean|R| / max|R| (**even quantity**)	26.3773 / 117.7287
`R_sum_lo`	Σ R of the low-energy band, λ > 250 nm	0.0000
`E_mean`	mean excitation energy (eV)	6.5169

Five points that must be disclosed:

1. **Units**: rotational strengths are treated in **cgs 10^{-40} erg·esu·cm/Gauss**; the exponent `10**-NN` in the original unit labels **differs from molecule to molecule** (median 5483, range 40–10926) ⇒ **it cannot be used as a scale factor**.
2. **The spectral window is narrow**: all molecule–transition λ values fall in **104–504 nm** (median 187 nm); only **55.4 %** of molecules have any transition with λ > 250 nm, and transitions with λ > 300 nm account for **1.03 %** ⇒ these 20 states essentially cover only the **far ultraviolet**, and **the Cotton-effect region of real ECD (250–400 nm) is almost never computed**. It remains usable as a "chiral pseudoscalar projection"; **for spectral prediction or absolute-configuration assignment one must add excited states**.
3. **Mapping onto CMRT**: matching by stereo-stripped canonical SMILES gives **9,939 / 10,117 = 98.2 %**; 4,825 (47.7 %) match the stereochemistry of `smi_S` and 5,116 (50.6 %) that of `smi_R`, with a **union of 100.0 %** (only 2 hit both S and R) ⇒ CMCDS does not mislabel stereochemistry; molecules falling on the R side must be **sign-flipped as a whole** when aligned, otherwise half of the pseudoscalars would have the wrong sign. After alignment the **directed coverage = 98.2 %**.
4. **Single-conformer approximation**: CMCDS uses the lowest-energy single conformer. This project has already measured that a single conformer is insufficient to support position-type vector triple products (sign-flip fraction only 0.363); R is a spectral quantity with low sensitivity, but it must still be declared.
5. **It is a genuine pseudoscalar**: the sign of `R_sum` **is not locked to CIP** (positive in 49.98 % of the 10,887 computed molecules, i.e. essentially balanced) — which is precisely why it is worth trying, and how it differs fundamentally from L1a/L1-CF.

S5.4 The downstream criterion shared by all three generations (fix the criterion first, then read the numbers)

Criterion: ① the ΔR^2 against the L0 baseline must be reported alongside the **SD of R^2 across folds**; ② "the signed columns contribute substantially" requires `signed − even > 0.015`.

generation	ΔR^2 (relative to L0, S2 / S2b / S3 / S4)	compared with the across-fold SD	signed vs even (within the same split)
L1a-FF	**+0.0416 / +0.0323 / +0.0382 / +0.0640** (`L0+D` arm)	**S2 and S4 exceed the SD** (S2b's +0.0323 < SD 0.0599; S3's +0.0382 < SD 0.3889)	**−0.010 ~ +0.015** (S2 +0.0146, S2b −0.0101, S3 +0.0054, S4 +0.0145) ⇒ **criterion (> 0.015): no substantial contribution**; and the direction is inconsistent (in S2b the even quantity is actually better)
L1-CF	**+0.0666 / +0.0615 / +0.0506 / +0.0888** (`L0+CF` arm)	**S2, S2b and S4 exceed the SD**; S3's +0.0506 < SD 0.3889 (CF reduces the S3 SD from 0.389 to **0.163**, i.e. actually more stable)	**+0.0010 ~ +0.0104** (S2 +0.0091, S2b +0.0061, S3 +0.0010, S4 +0.0104) ⇒ **criterion (> 0.015): no substantial contribution**
L1b	**−0.0221 / +0.0220 / +0.0100 / −0.0043** (`L0+R_sgn` arm)	**does not exceed** (negative or very low in all four splits); added to the best arm, S2 −0.0066, S2b −0.0028, S4 −0.0025 (**worse in 3 of 4 splits**), S3 +0.0025	**signed is worse or unchanged** (S2 −0.0221 vs −0.0092; S4 −0.0043 vs −0.0182, slightly better; S2b/S3 slightly worse)

Across-fold R^2 SD (L0 baseline, four splits): **0.0187 / 0.0599 / 0.3889 / 0.0187** (S2 / S2b / S3 / S4) — any ΔR^2 must be reported alongside this row.

Reading (no overclaiming):

- **The descriptor design direction is right**: L1-CF beats L1a-FF in 4/4 splits (Δ = +0.025 / +0.029 / +0.012 / +0.025).
- **But the gain comes from "describing shape in a local chiral frame", not from chirality**: the gain of the signed columns over the even-quantity columns is **in every case below the pre-registered 0.015 threshold**.
- **S2b and S3 remain negative**; the distance to $R^2 \geq 0.6$ **is still very large** (under S2c, even giving away each paper's mean and scale for free reaches only 0.4393).

S6. Models, hyperparameters and tuning protocol

Software environment: Python **3.14.3**; RDKit **2026.03.6**; scikit-learn **1.9.0**; NumPy **2.4.6**; pandas **3.0.3**; SciPy **1.18.0**. Fixed random seed **`SEED = 42`** (point-wise shuffling in S1, ECFP4 sampling); geometric seed for the L1 descriptors **`0xC1A1`**.

purpose	model	hyperparameters (fixed; no per-fold tuning)
default regression (when data are sufficient)	`HistGradientBoostingRegressor`	200–300 iterations; `learning_rate = 0.06`; `min_samples_leaf = 20`
default classification (EMO)	`HistGradientBoostingClassifier`	`max_iter = 200`; `learning_rate =`

purpose	model	hyperparameters (fixed; no per-fold tuning)
		`0.06`; `min_samples_leaf = 20`
small-sample regression / within-laboratory contrast (Section 3.3)	standardized `Ridge`	**fixed `alpha = 10` (no cross-validated choice of alpha)**
small-sample classification	`LogisticRegressionCV`	defaults; like Ridge, a model class estimable at small sample size
CSP extrapolation (S3′)	same as default	167-dimensional self-built CSP descriptors

Why hyperparameters are fixed rather than tuned (a methodological position):

1. The claim of the paper is **about data structure**, not about model capacity; tuning would only make the "failure" harder to explain.
2. The contrast arms **must share the same model capacity**, otherwise ΔR^2 cannot be attributed.
3. **Cross-validation is itself unstable at small sample sizes**: the within-laboratory training half in Section 3.3 has only ~15 rows, and a GBM with `min_samples_leaf = 20` **cannot make a single split** (it would **artificially force the "within-laboratory" result to 0**) ⇒ that setting always uses Ridge; GBM serves only as a **secondary contrast** for the 12 papers with `n ≥ 60` (median per-paper r +0.266, r>0 in 50%).
4. **R^2 is unusable on low-variance splits**: outputs of order −1e15 were observed (in `within_paper_split.csv`, **23 papers** have $|R^2| > 100$, the most extreme being −2.48e18) ⇒ for such comparisons the **primary metric is always the scale-free per-paper Pearson r**.
5. **The t_o porosity is an externally supplied constant** (reference φ = 0.62) and is not fitted; its sensitivity is reported separately (S10 item 7).

Negative controls and power (throughout): every quantitative conclusion is paired with a shuffled-label control run through the same pipeline (on these data `|R²| ≤ 0.064`, EMO ≈ 50.2%); every null result is paired with a power curve (S8.3).

S7. Full tables of per-fold / per-paper / per-condition / per-column results

⚠ **Every table in this section was generated by the released scripts reading `data/curated/*.csv` directly** (these scripts only tabulate and recompute nothing); no number was entered by hand. Column names match the CSV column names, so that reviewers can check every cell.

S7-A Split index files (per-fold sizes, tabulated directly from `data/curated/splits/*.csv`)

file	rows	cols	folds	$size_{min}$	$size_{med}$	$size_{max}$	kind
S1_random.csv	10525	2	2	2105	5262.5	8420	per-fold
S2_$group_{dup}$.csv	10525	2	5	2105	2105.0	2105	per-fold
S2b_{leave}_paper.csv	10525	3	10	1052	1052.5	1053	per-fold
S2c_{leave}_doi.csv	10525	3	530	1	18.0	120	per-fold
S3_$leave_{column}$.csv	10525	3	25	13	247.0	2216	per-fold
S3p_{factor}.csv	17046	4	—	—	—	—	long table (task/role, no fold column)
S4_murcko.csv	10525	3	5	2105	2105.0	2105	per-fold
S4b_{ecfp}_class.csv	10525	3	5	2105	2105.0	2105	per-fold

Note: the S3 split file has 25 folds, but the per-fold statistics table (S7-C) contains only 23 — the remaining 2 folds have too few samples per fold and are filtered out by `fold_stats(min_n)`, so they do not enter the per-fold statistics (they do enter the global OOF metrics). Note: `size_med` is the **median fold size**; a .5 is normal (median of an even number of folds: 2 folds in S1 → 5262.5; 10 folds in S2b → 1052.5).

S7-B EMO by protocol (same classifier / same L0 features; sources `s2c_emo.csv` + `s3_emo.csv`)

protocol	folds	n	emo	$base_{global}$	$base_{perfold}$	$gain_{pp}$
S1 random (point-wise 80/20)	2	10525	66.13%	50.67%	50.87%	+15.46
S2 grouped (molecule, column)	5	10525	71.79%	50.67%	51.00%	+21.12
S2b leave-a-batch-of-papers (~53 papers/fold)	10	10525	53.94%	50.67%	52.87%	+3.27
S2c true leave-one-paper-out	530	10525	52.85%	50.67%	76.55%	+2.18
S4 leave-one-scaffold-out	5	10525	58.71%	50.67%	52.85%	+8.04
S3 leave-one-column-out	25	—	56.46%	50.67%	—	+5.79

protocol	folds	n	emo	$base_{global}$	$base_{perfold}$	$gain_{pp}$
negative control (shuffled labels, S2c protocol, 3 runs)	—	—	50.17% ± 0.22%	—	—	—
global majority-class baseline	—	—	50.67%	—	—	—

S7-C Full per-fold failure-mode decomposition (`failure_decomposition.csv`, 86 rows = 4 splits × 2 arms × folds)

Columns: `y_mean/p_mean` = within-fold target/prediction mean; `bias` = within-fold offset; `slope` = within-fold OLS slope; `r/rho` = Pearson/Spearman; `r2/r2_off/r2_aff` = raw / oracle de-offset / oracle rescaled.

S2 grouped (molecule × column) — L0 (5 folds)

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
1	2105	-0.0009	0.4213	+0.0011	0.2007	-0.0020	1.1196	+0.5335	+0.5149	+0.2813	+0.2813	+0.2846
2	2105	+0.0072	0.3999	+0.0018	0.1991	+0.0055	0.9619	+0.4789	+0.5012	+0.2288	+0.2290	+0.2293
3	2105	+0.0056	0.4043	-0.0042	0.2049	+0.0098	1.0620	+0.5381	+0.5157	+0.2880	+0.2886	+0.2895
4	2105	-0.0068	0.4017	+0.0007	0.2033	-0.0076	1.0056	+0.5088	+0.4641	+0.2585	+0.2589	+0.2589
5	2105	+0.0060	0.4320	+0.0073	0.2018	-0.0013	1.1522	+0.5382	+0.5473	+0.2846	+0.2846	+0.2897

S2 grouped (molecule × column) — L0+CF+FF (5 folds)

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
1	2073	-0.0017	0.4225	-0.0025	0.2253	+0.0007	1.1463	+0.6114	+0.5930	+0.3677	+0.3677	+0.3738
2	2079	+0.0058	0.3993	+0.0019	0.2293	+0.0039	0.9863	+0.5664	+0.5991	+0.3207	+0.3208	+0.3208
3	2069	+0.0040	0.4052	+0.0005	0.2348	+0.0035	1.0378	+0.6015	+0.5816	+0.3612	+0.3613	+0.3618
4	2074	-0.0086	0.4026	-0.0005	0.2281	-0.0081	1.0269	+0.5817	+0.5554	+0.3378	+0.3382	+0.3384
5	2068	+0.0054	0.4308	+0.0043	0.2377	+0.0011	1.1093	+0.6120	+0.6162	+0.3709	+0.3709	+0.3745

S2b leave-one-paper-out — L0 (10 folds)

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
1	1053	+0.0151	0.4031	+0.0156	0.1552	-0.0005	0.2311	+0.0890	+0.0754	-0.0798	-0.0798	+0.0079
2	1053	+0.0111	0.4050	-0.0030	0.1507	+0.0140	0.0532	+0.0198	+0.0647	-0.1250	-0.1238	+0.0004

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
3	1053	+0.0087	0.4073	+0.0187	0.1348	-0.0100	0.4137	+0.1369	+0.1179	-0.0195	-0.0189	+0.0187
4	1053	-0.0022	0.3999	-0.0179	0.1701	+0.0158	0.3230	+0.1374	+0.2065	-0.0656	-0.0640	+0.0189
5	1053	-0.0059	0.4163	+0.0219	0.1394	-0.0278	-0.0929	-0.0311	-0.0249	-0.1374	-0.1329	+0.0010
6	1052	-0.0468	0.3841	+0.0147	0.1396	-0.0615	-0.1541	-0.0560	-0.0526	-0.1985	-0.1729	+0.0031
7	1052	-0.0221	0.4439	+0.0060	0.1297	-0.0281	0.2911	+0.0850	+0.0694	-0.0397	-0.0357	+0.0072
8	1052	+0.0455	0.3949	+0.0455	0.1267	+0.0000	0.2930	+0.0940	+0.0728	-0.0426	-0.0426	+0.0088
9	1052	-0.0198	0.4707	+0.0314	0.1610	-0.0511	-0.3474	-0.1189	-0.0338	-0.2102	-0.1984	+0.0141
10	1052	+0.0384	0.3785	+0.0429	0.1382	-0.0044	0.1259	+0.0460	+0.0746	-0.0999	-0.0997	+0.0021

S2b leave-one-paper-out — L0+CF+FF (10 folds)

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
1	1053	+0.0151	0.4031	+0.0040	0.1772	+0.0111	0.3441	+0.1513	+0.1791	-0.0610	-0.0603	+0.0229
2	1052	+0.0100	0.4036	-0.0165	0.1570	+0.0265	0.3668	+0.1427	+0.1801	-0.0446	-0.0403	+0.0204
3	1053	+0.0087	0.4073	-0.0192	0.1221	+0.0279	0.3662	+0.1098	+0.0391	-0.0287	-0.0240	+0.0120
4	1024	-0.0075	0.3964	-0.0244	0.1431	+0.0170	0.4875	+0.1760	+0.1775	-0.0051	-0.0033	+0.0310
5	965	-0.0060	0.4183	+0.0111	0.1321	-0.0171	0.0291	+0.0092	+0.0400	-0.0957	-0.0940	+0.0001
6	1052	-0.0468	0.3841	-0.0009	0.1375	-0.0459	0.3079	+0.1102	+0.1103	-0.0635	-0.0492	+0.0121
7	1052	-0.0221	0.4439	+0.0032	0.1494	-0.0253	0.6454	+0.2171	+0.2143	+0.0297	+0.0329	+0.0472
8	1052	+0.0455	0.3949	+0.0125	0.1792	+0.0330	0.3787	+0.1718	+0.1472	-0.0570	-0.0500	+0.0295
9	1049	-0.0202	0.4710	+0.0201	0.1270	-0.0403	0.2062	+0.0556	+0.0265	-0.0501	-0.0427	+0.0031
10	1011	+0.0332	0.3829	+0.0543	0.1352	-0.0211	0.2125	+0.0751	+0.0373	-0.0748	-0.0717	+0.0056

S3 leave-one-column-out — L0 (23 folds)

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
1	265	-0.0531	0.4932	+0.0414	0.1668	-0.0945	-0.2902	-0.0982	-0.0144	-0.2175	-0.1808	+0.0096
2	265	+0.1051	0.4143	+0.0407	0.1475	+0.0644	0.0110	+0.0039	+0.1340	-0.1481	-0.1239	+0.0000
3	2216	+0.0345	0.4354	+0.0121	0.1643	+0.0223	0.3296	+0.1244	+0.1507	-0.0512	-0.0485	+0.0155
4	152	-0.0112	0.3827	+0.0356	0.1045	-0.0468	0.2925	+0.0799	-0.0009	-0.0459	-0.0309	+0.0064
5	487	-0.0621	0.3831	-0.0060	0.1458	-0.0561	0.2114	+0.0804	+0.0148	-0.1050	-0.0836	+0.0065
6	156	+0.3180	0.2608	+0.0191	0.1211	+0.2989	-0.2698	-0.1253	+0.1115	-1.6453	-0.3320	+0.0157
7	714	-0.0060	0.3982	-0.0254	0.1678	+0.0193	0.4957	+0.2089	+0.3022	-0.0039	-0.0015	+0.0437
10	269	-0.1343	0.3392	-0.0297	0.1617	-0.1046	0.3356	+0.1600	+0.1175	-0.1697	-0.0747	+0.0256
11	857	-0.0021	0.3844	-0.0041	0.1751	+0.0020	0.2088	+0.0951	+0.0926	-0.1208	-0.1208	+0.0090

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
12	412	+0.0852	0.3264	+0.0213	0.1146	+0.0639	0.7303	+0.2563	+0.2415	+0.0185	+0.0567	+0.0657
13	98	-0.0737	0.3278	+0.0110	0.1395	-0.0847	-0.1945	-0.0828	-0.0291	-0.3183	-0.2515	+0.0068
14	247	+0.0520	0.3792	+0.0106	0.1475	+0.0413	0.0333	+0.0129	+0.0348	-0.1530	-0.1412	+0.0002
15	388	-0.0513	0.3407	+0.0519	0.1409	-0.1032	0.3709	+0.1533	+0.1875	-0.1359	-0.0441	+0.0235
16	201	-0.1424	0.3169	+0.0087	0.1155	-0.1511	0.6652	+0.2425	+0.2104	-0.1836	+0.0439	+0.0588
17	77	+0.0170	0.6237	+0.0297	0.1335	-0.0127	1.4079	+0.3014	+0.3516	+0.0828	+0.0832	+0.0908
18	111	+0.0184	0.2773	+0.0008	0.1074	+0.0176	0.8090	+0.3132	+0.2306	+0.0886	+0.0926	+0.0981
19	165	+0.0954	0.3646	+0.0031	0.1256	+0.0923	0.8153	+0.2808	+0.3877	+0.0107	+0.0748	+0.0788
20	338	+0.0471	0.4754	+0.0043	0.1513	+0.0428	0.8078	+0.2571	+0.2500	+0.0542	+0.0624	+0.0661
21	1966	-0.0435	0.4123	-0.0122	0.1558	-0.0313	0.2000	+0.0756	+0.1085	-0.0914	-0.0856	+0.0057
22	94	+0.0954	0.6471	+0.0006	0.0831	+0.0948	0.5517	+0.0708	+0.0721	-0.0198	+0.0017	+0.0050
23	123	-0.0133	0.3309	-0.0323	0.1384	+0.0190	0.3628	+0.1518	+0.1476	-0.0513	-0.0480	+0.0230
24	788	+0.0080	0.4106	-0.0026	0.1263	+0.0105	0.4477	+0.1377	+0.1717	-0.0105	-0.0099	+0.0190
25	104	-0.0127	0.3539	+0.0121	0.1227	-0.0248	-0.0093	-0.0032	+0.0818	-0.1274	-0.1225	+0.0000

S3 leave-one-column-out — L0+CF+FF (23 folds)

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
1	264	-0.0544	0.4937	+0.0010	0.1680	-0.0555	0.6524	+0.2220	+0.2207	+0.0227	+0.0353	+0.0493
2	265	+0.1051	0.4143	+0.0729	0.1603	+0.0322	0.2784	+0.1077	+0.2200	-0.0724	-0.0664	+0.0116
3	2155	+0.0319	0.4383	-0.0047	0.1993	+0.0366	0.4018	+0.1826	+0.2093	-0.0476	-0.0406	+0.0334
4	148	-0.0234	0.3506	+0.0219	0.1188	-0.0452	0.6774	+0.2296	+0.2062	+0.0241	+0.0407	+0.0527
5	485	-0.0650	0.3809	-0.0402	0.1376	-0.0248	0.7151	+0.2583	+0.1288	+0.0519	+0.0561	+0.0667
6	156	+0.3180	0.2608	+0.1260	0.1170	+0.1919	0.9255	+0.4151	+0.3593	-0.3705	+0.1712	+0.1723
7	712	-0.0057	0.3987	-0.0108	0.1735	+0.0052	0.5057	+0.2201	+0.3231	+0.0020	+0.0022	+0.0485
10	267	-0.1351	0.3400	-0.0494	0.1805	-0.0857	0.3043	+0.1615	+0.2149	-0.1739	-0.1103	+0.0261
11	844	-0.0034	0.3866	+0.0299	0.1843	-0.0333	0.2704	+0.1289	+0.1676	-0.1117	-0.1043	+0.0166
12	412	+0.0852	0.3264	+0.0092	0.1296	+0.0760	0.4536	+0.1801	+0.1772	-0.0688	-0.0146	+0.0324
13	98	-0.0737	0.3278	+0.0070	0.1738	-0.0807	0.1270	+0.0674	-0.0386	-0.2702	-0.2097	+0.0045
14	243	+0.0537	0.3818	-0.0010	0.1534	+0.0547	0.3351	+0.1346	+0.1452	-0.0738	-0.0532	+0.0181
15	386	-0.0512	0.3414	+0.0281	0.1352	-0.0793	0.3576	+0.1416	+0.0990	-0.0986	-0.0447	+0.0201
16	201	-0.1424	0.3169	-0.0242	0.1622	-0.1183	0.8652	+0.4430	+0.3579	+0.0521	+0.1915	+0.1962
17	77	+0.0170	0.6237	+0.1016	0.1439	-0.0846	1.6387	+0.3781	+0.4965	+0.1028	+0.1212	+0.1429
18	111	+0.0184	0.2773	+0.0233	0.1136	-0.0050	0.9144	+0.3744	+0.3795	+0.1387	+0.1390	+0.1402
19	165	+0.0954	0.3646	+0.0067	0.1529	+0.0887	0.3936	+0.1651	+0.2146	-0.0967	-0.0374	+0.0273

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
20	338	+0.0471	0.4754	+0.0031	0.1701	+0.0440	0.4320	+0.1545	+0.0956	-0.0260	-0.0174	+0.0239
21	1939	-0.0469	0.4135	+0.0042	0.1804	-0.0511	0.2483	+0.1083	+0.1231	-0.1111	-0.0958	+0.0117
22	91	+0.0763	0.6439	-0.0056	0.0777	+0.0820	1.1224	+0.1355	+0.2376	+0.0019	+0.0181	+0.0184
23	123	-0.0133	0.3309	-0.0331	0.1220	+0.0199	0.6661	+0.2455	+0.2037	+0.0415	+0.0451	+0.0603
24	748	+0.0171	0.4086	+0.0162	0.1416	+0.0009	0.6228	+0.2158	+0.2418	+0.0295	+0.0295	+0.0466
25	104	-0.0127	0.3539	+0.0078	0.0992	-0.0204	-0.5375	-0.1507	-0.0450	-0.1664	-0.1631	+0.0227

S4 leave-one-scaffold-out — L0 (5 folds)

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
1	2105	+0.0456	0.3854	-0.0232	0.1416	+0.0688	0.7552	+0.2775	+0.2712	+0.0371	+0.0689	+0.0770
2	2105	+0.0090	0.4120	+0.0186	0.1555	-0.0096	0.8199	+0.3094	+0.2692	+0.0905	+0.0911	+0.0957
3	2105	-0.0101	0.4252	-0.0188	0.1715	+0.0087	0.7016	+0.2831	+0.2313	+0.0652	+0.0656	+0.0801
4	2105	-0.0161	0.4113	-0.0035	0.1521	-0.0126	0.6980	+0.2582	+0.2639	+0.0532	+0.0542	+0.0666
5	2105	-0.0174	0.4218	+0.0037	0.1617	-0.0210	0.7060	+0.2706	+0.2534	+0.0581	+0.0605	+0.0732

S4 leave-one-scaffold-out — L0+CF+FF (5 folds)

fold	n	y_{mean}	y_{sd}	p_{mean}	p_{sd}	bias	slope	r	rho	r2	r2_off	r2_aff
1	2061	+0.0420	0.3841	+0.0120	0.1576	+0.0299	0.8214	+0.3370	+0.3321	+0.1021	+0.1082	+0.1136
2	2092	+0.0089	0.4121	+0.0123	0.1734	-0.0034	1.0335	+0.4349	+0.4084	+0.1888	+0.1889	+0.1891
3	2086	-0.0115	0.4261	-0.0134	0.1836	+0.0018	0.9756	+0.4205	+0.3624	+0.1767	+0.1767	+0.1768
4	2079	-0.0173	0.4133	-0.0046	0.1730	-0.0128	0.9551	+0.3998	+0.3973	+0.1585	+0.1594	+0.1598
5	2045	-0.0171	0.4213	+0.0018	0.1880	-0.0190	0.9831	+0.4386	+0.4041	+0.1903	+0.1923	+0.1924

S7-D S2c (true leave-one-paper-out) OOF summary and the `R²_aff ≡ η²` invariant (`leave_one_doi_summary.csv`)

⚠ **Convention**: the global R^2 in this table is not "the OOF R^2 of all 10,525 rows" but one **rebuilt with fold-size weighting** by `scripts/13.summarize()`, including only folds with $n \geq 20$ within the fold ⇒ it covers **10193** rows / **428** folds (428 papers). The complete S2c descriptor-arm evaluation subset is **10,363 rows / 9,968 molecules / 523 papers** (a molecule must fall within the coverage of both L1-CF and L1a-FF); the EMO classification arm uses **all 10,525 rows / 530 papers**. The S2c split file `S2c_leave_doi.csv` itself contains **530 folds** (one paper = one DOI = one fold). The "523 folds" in the main text = **the number of papers evaluated**, not the number of folds in the split file — the two must be reported together.

split	arm	n	n_{fold}	r2	r2_off	r2_aff	b_{iqr}	b_{med}	r_{med}	rho_{med}	$frac_{r}_gt0$	$frac_{aff}_gt0$	$frac_{off}_gt0$
L0	L0	10193	428	-0.1073	+0.3337	+0.4393	+0.3163	-0.0032	+0.0459	+0.0394	56.1%	100.0%	33.6%
L0+CF+FF	L0+CF+FF	10193	428	-0.0477	+0.3446	+0.4419	+0.3045	-0.0080	+0.0119	+0.0173	51.9%	100.0%	29.7%

η^2(doi) = 0.3887 (523 folds / n=10,363); `R²_aff(L0) − η² = 0.4393 − 0.3887 = +0.0506`. ⇒ **even when each paper's own mean and scale are handed over for free, R^2 only reaches 0.4393 < 0.6**; the within-fold ranking ability actually contributed by the descriptors is only **+0.05**.

S7-E Across-paper (change of laboratory) learning curve (`learning_curve_papers.csv`, k = number of training papers)

k	n_{train}	r2	r2_sd	r2_aff	r_{med}	$frac_{r}_gt0$	emo	emo_{base}
1	14.1	-1678.6326	+2769.0836	+0.0091	-0.0468	44.3%	49.81%	52.63%
2	32.4	-1.4900	+2.1480	+0.0283	+0.0102	51.2%	47.05%	52.85%
5	110.4	-1.1157	+0.7252	+0.0045	+0.0357	53.7%	49.16%	53.58%
10	178.1	-0.5212	+0.2473	+0.0054	+0.0093	49.2%	47.25%	52.91%
20	403.1	-0.3523	+0.3328	+0.0098	+0.0332	56.0%	51.36%	56.25%
50	994.2	-0.1870	+0.1342	+0.0170	+0.0318	53.2%	51.20%	54.18%
100	1990.0	-0.1149	+0.0454	+0.0055	+0.0053	52.0%	49.17%	53.54%
250	5037.0	-0.0498	+0.0469	+0.0088	+0.0009	49.4%	49.90%	53.06%

`r2_aff` is recalibrated **en bloc** (the 40 test papers pooled together); it is not the same estimator as the **per-fold** `R²_aff` of `scripts/15`, and the two **must not be compared side by side**. Known small leakage: 297 molecules (2.9%) occur in ≥2 papers, whereas this setting only guarantees **non-overlapping papers** ⇒ this column's R^2 is slightly **optimistic** (the direction is unfavourable to the conclusion).

S7-F Within-laboratory (within a single paper) full per-paper results (`within_paper_split.csv`, 170 papers / 6,325 rows)

Each paper is split 50/50 by molecule, repeated 8 times and averaged (only papers with n ≥ 25 are included).

doi	n	r2	r	r2_aff	emo	emo_{base}
10.1002/CJOC.201800088	32	-0.3517	+0.0708	+0.0279	66.64%	53.35%
10.1002/adsc.201600353	30	+0.0840	+0.5016	+0.2766	100.00%	100.00%
10.1002/adsc.201700666	35	n/a (near-zero variance)	+0.2319	+0.0844	97.22%	97.22%
10.1002/adsc.201800839	25	-0.2403	+0.3213	+0.1364	73.08%	71.15%
10.1002/anie.201105812	28	n/a (near-zero variance)	+0.1655	+0.2640	66.96%	61.61%
10.1002/anie.201300188	47	-0.3454	+0.1086	+0.0285	46.88%	55.73%
10.1002/anie.201302861	32	-0.4396	+0.3017	+0.1284	75.78%	77.34%

doi	n	r2	r	r2_aff	emo	emo_{base}
10.1002/anie.201303247	26	n/a (near-zero variance)	-0.0575	+0.0132	38.46%	59.62%
10.1002/anie.201305861	27	-0.6093	+0.1756	+0.0724	56.25%	62.50%
10.1002/anie.201412132	25	-1.8371	+0.2419	+0.0868	68.27%	73.08%
10.1002/anie.201501762	25	-2.2362	-0.2642	+0.1710	92.31%	98.08%
10.1002/anie.201504483	27	-0.4773	+0.5633	+0.3265	86.61%	86.61%
10.1002/anie.201506701	28	-0.9832	-0.1931	+0.0961	36.61%	61.61%
10.1002/anie.201508108	31	-0.3297	+0.1736	+0.0561	89.84%	90.62%
10.1002/anie.201601844	37	-0.3755	-0.0820	+0.0779	43.42%	56.58%
10.1002/anie.201605530	30	-1.0504	+0.1167	+0.0408	42.97%	66.04%
10.1002/anie.201606571	26	-0.0140	+0.4512	+0.2558	71.15%	78.85%
10.1002/anie.201612208	26	n/a (near-zero variance)	+0.0133	+0.0617	100.00%	100.00%
10.1002/anie.201701947	42	-0.2538	+0.3154	+0.1170	73.81%	76.79%
10.1002/anie.201702925	28	n/a (near-zero variance)	+0.1695	+0.0621	82.14%	85.71%
10.1002/anie.201712395	33	-0.3067	+0.1240	+0.0652	47.79%	53.68%
10.1002/anie.201804197	49	n/a (near-zero variance)	+0.0217	+0.0103	52.50%	57.50%
10.1002/anie.201804315	27	-0.6164	-0.0357	+0.0606	100.00%	100.00%
10.1002/anie.201806372	34	-0.3610	+0.0245	+0.0160	79.41%	81.62%
10.1002/anie.201810879	26	n/a (near-zero variance)	+0.2832	+0.1155	85.58%	90.38%
10.1002/anie.201812426	28	-0.9423	+0.2909	+0.1200	82.14%	82.14%
10.1002/anie.201902576	36	+0.4257	+0.7258	+0.5332	95.83%	81.94%
10.1002/anie.201904805	34	-0.0684	+0.4236	+0.1828	67.65%	57.35%
10.1002/anie.201905263	30	-39.7538	+0.2580	+0.2630	100.00%	100.00%
10.1002/anie.201907840	81	-0.0939	+0.2219	+0.0560	49.70%	56.10%
10.1002/anie.201909919	32	-0.1806	+0.3106	+0.1131	70.31%	55.47%
10.1002/anie.201909928	33	-13.4902	+0.3482	+0.1441	91.91%	94.12%
10.1002/anie.201910183	29	-0.5150	+0.4091	+0.1952	97.50%	97.50%
10.1002/anie.201912160	43	-0.1935	+0.3174	+0.1517	74.43%	57.95%
10.1002/anie.201912408	87	-0.2566	+0.1178	+0.0414	77.84%	79.26%
10.1002/anie.201912926	32	-0.6418	+0.0348	+0.0365	66.41%	73.44%
10.1002/anie.201913367	106	+0.4981	+0.7388	+0.5485	88.92%	60.61%
10.1002/anie.201915459	31	-1.3816	+0.0211	+0.0117	84.38%	88.28%
10.1002/anie.201916534	27	-1.0983	+0.2857	+0.1276	73.21%	79.46%
10.1002/anie.202004485	33	-0.8216	+0.1664	+0.0558	64.71%	74.26%
10.1002/anie.202004504	29	-0.0317	+0.3009	+0.1604	90.83%	89.17%

doi	n	r2	r	r2_aff	emo	emo_{base}
10.1002/anie.202005313	25	n/a (near-zero variance)	-0.0811	+0.2738	93.27%	93.27%
10.1002/anie.202006317	55	-0.2864	+0.0850	+0.0607	70.09%	73.66%
10.1002/anie.202008262	42	-0.2498	+0.2238	+0.0832	79.87%	79.87%
10.1002/anie.202009358	39	-0.6937	+0.1646	+0.0418	73.75%	76.88%
10.1002/anie.202009527	46	-0.3309	+0.1727	+0.0721	87.91%	88.43%
10.1002/anie.202010164	35	-0.0279	+0.5512	+0.3164	72.22%	71.53%
10.1002/anie.202012485	32	+0.3124	+0.6767	+0.4676	93.75%	86.72%
10.1002/anie.202016506	43	+0.1422	+0.5518	+0.3230	94.89%	68.75%
10.1002/anie.202105452	46	-0.4112	+0.1946	+0.0606	77.72%	79.35%
10.1002/anie.202106109	46	-0.1409	+0.3113	+0.1230	72.28%	71.74%
10.1002/anie.202106566	25	n/a (near-zero variance)	+0.3661	+0.3566	95.19%	95.19%
10.1002/anie.202107909	35	-0.2402	+0.1747	+0.0656	63.19%	64.58%
10.1002/anie.202110084	69	-0.3979	+0.0385	+0.0452	98.93%	98.93%
10.1002/anie.202110233	59	-0.2521	+0.1789	+0.0592	58.33%	58.75%
10.1002/anie.202110286	28	+0.4121	+0.7839	+0.6369	100.00%	100.00%
10.1002/anie.202111957	28	+0.0959	+0.5217	+0.2956	76.79%	77.68%
10.1002/anie.202112860	34	-1.3186	-0.0024	+0.0233	62.50%	71.32%
10.1002/anie.202202552	42	-0.2921	+0.2100	+0.0557	81.55%	81.55%
10.1002/chem.201405830	30	-0.2007	+0.2587	+0.0875	60.57%	62.97%
10.1002/chem.201501481	34	+0.0681	+0.5167	+0.3015	62.50%	58.82%
10.1002/chem.201704760	30	-0.0864	+0.4472	+0.2487	85.00%	87.50%
10.1002/chem.202004613	61	-0.1623	+0.1838	+0.0450	61.29%	60.08%
10.1002/chem.202100829	44	-0.2707	+0.3774	+0.1670	82.95%	84.66%
10.1002/cjoc.201900533	29	n/a (near-zero variance)	-0.0284	+0.0246	72.50%	74.17%
10.1002/cjoc.202100538	35	+0.0499	+0.4053	+0.2186	79.17%	76.39%
10.1002/cjoc.202100809	31	-1.0111	+0.2602	+0.0914	90.62%	90.62%
10.1002/ntls.10021	58	n/a (near-zero variance)	+0.5576	+0.3347	82.76%	59.48%
10.1016/j.chempr.2017.10.008	43	-0.5460	+0.0980	+0.0549	76.70%	80.68%
10.1016/j.chempr.2018.03.010	33	n/a (near-zero variance)	+0.0263	+0.0610	49.26%	55.88%

doi	n	r2	r	r2_aff	emo	emo_{base}
10.1016/ j.isci.2020.101138	52	-0.1186	+0.1917	+0.1413	50.05%	55.93%
10.1021/ acs.organomet.9b00366	29	-0.3568	+0.1516	+0.1185	60.83%	63.33%
10.1021/ acs.orglett.0c00461	37	+0.0984	+0.5995	+0.3692	80.92%	85.53%
10.1021/ acs.orglett.0c00669	31	-0.4951	-0.1688	+0.0310	49.22%	57.03%
10.1021/ acs.orglett.0c00748	27	-1.5678	-0.1085	+0.0443	66.96%	72.32%
10.1021/ acs.orglett.0c01486	44	-0.1970	+0.1152	+0.1003	83.52%	87.50%
10.1021/ acs.orglett.0c01626	26	-0.1999	+0.2357	+0.1911	50.96%	57.69%
10.1021/ acs.orglett.0c02487	25	-0.8181	+0.0645	+0.0317	55.77%	64.42%
10.1021/ acs.orglett.0c03466	30	n/a (near-zero variance)	+0.1090	+0.0897	89.17%	90.00%
10.1021/ acs.orglett.1c00974	32	n/a (near-zero variance)	+0.2878	+0.0890	100.00%	100.00%
10.1021/ acs.orglett.1c01701	44	-0.0181	+0.3709	+0.1587	75.57%	69.89%
10.1021/ acs.orglett.1c01792	34	-0.1290	+0.2607	+0.0774	94.12%	94.12%
10.1021/ acs.orglett.1c02433	30	-0.9234	-0.0680	+0.1700	90.83%	89.17%
10.1021/ acs.orglett.1c02918	41	-0.4441	+0.0068	+0.0134	77.38%	79.17%
10.1021/ acs.orglett.2c00213	32	-33.9659	+0.1809	+0.0896	84.38%	89.06%
10.1021/ acs.orglett.5b03072	31	-3.5848	+0.1289	+0.0567	72.66%	74.22%
10.1021/ acs.orglett.6b01737	25	+0.2167	+0.6299	+0.4133	73.08%	64.42%
10.1021/ acs.orglett.6b02095	35	-0.7738	-0.1155	+0.1044	87.87%	89.34%
10.1021/ acs.orglett.6b03161	26	+0.3444	+0.7061	+0.5181	69.23%	70.19%

doi	n	r2	r	r2_aff	emo	emo_{base}
10.1021/acs.orglett.7b00128	26	-1.9428	-0.2405	+0.1113	73.08%	82.69%
10.1021/acs.orglett.7b01136	33	-0.5344	+0.4607	+0.2188	85.29%	84.56%
10.1021/acs.orglett.7b01904	38	n/a (near-zero variance)	+0.1084	+0.0530	57.24%	63.16%
10.1021/acs.orglett.7b03650	32	-0.2835	+0.0998	+0.0957	53.12%	55.47%
10.1021/acs.orglett.8b00740	25	-0.3741	+0.1298	+0.0824	80.77%	80.77%
10.1021/acs.orglett.8b01029	38	+0.2039	+0.6150	+0.3886	66.45%	69.08%
10.1021/acs.orglett.8b02591	27	-0.8857	+0.2595	+0.1279	90.18%	91.07%
10.1021/acs.orglett.8b02902	36	-31.8037	+0.3721	+0.1708	84.03%	88.89%
10.1021/acs.orglett.9b00216	28	-0.6847	-0.1737	+0.0843	98.21%	98.21%
10.1021/acs.orglett.9b01290	25	-0.0952	+0.4212	+0.2378	91.35%	91.35%
10.1021/acs.orglett.9b01514	34	-0.3435	+0.4508	+0.2091	96.32%	96.32%
10.1021/acs.orglett.9b03452	35	-0.2510	+0.0727	+0.0277	59.72%	64.75%
10.1021/acs.orglett.9b03685	26	-5.9506	-0.3184	+0.1391	39.42%	64.42%
10.1021/acs.orglett.9b04244	29	-0.8447	+0.1462	+0.0682	64.17%	70.83%
10.1021/acscatal.0c00712	31	-0.3367	+0.3371	+0.1348	67.19%	60.16%
10.1021/acscatal.0c04206	32	-0.3457	+0.2339	+0.0858	46.88%	58.59%
10.1021/acscatal.1c04541	35	-0.6220	-0.0715	+0.0221	58.33%	55.56%
10.1021/acscatal.1c05008	50	-3.9834	-0.0157	+0.0264	88.50%	91.00%
10.1021/acscatal.7b00030	26	-0.7285	-0.0159	+0.1245	76.92%	78.85%

doi	n	r2	r	r2_aff	emo	emo_{base}
10.1021/acscatal.8b02543	36	-0.1953	+0.1884	+0.0705	75.69%	75.69%
10.1021/acscatal.9b02080	36	n/a (near-zero variance)	+0.0611	+0.0825	100.00%	100.00%
10.1021/acscatal.9b02667	25	-0.0800	+0.5257	+0.2954	75.00%	78.85%
10.1021/acscatal.9b03163	40	+0.1446	+0.4863	+0.2636	68.12%	72.50%
10.1021/ja510980d	33	-0.2432	+0.1967	+0.0419	54.41%	60.29%
10.1021/jacs.0c03130	73	-0.4935	+0.2098	+0.0528	70.95%	76.01%
10.1021/jacs.0c07210	38	-0.8210	-0.3135	+0.1203	66.45%	72.37%
10.1021/jacs.0c09125	45	-0.1163	+0.3287	+0.1267	64.07%	67.81%
10.1021/jacs.1c04016	30	-0.9632	-0.1633	+0.0543	57.50%	61.67%
10.1021/jacs.1c04367	49	-0.5025	+0.0541	+0.0222	77.00%	76.50%
10.1021/jacs.1c06535	63	-0.4331	+0.2096	+0.0527	96.88%	96.88%
10.1021/jacs.1c13384	34	-2.8608	-0.1421	+0.0331	78.68%	81.62%
10.1021/jacs.2c00957	86	+0.0256	+0.4215	+0.1844	70.64%	52.62%
10.1021/jacs.5b09117	35	-0.8873	-0.1820	+0.0483	91.67%	92.36%
10.1021/jacs.6b00386	41	-0.0578	+0.3680	+0.1795	70.24%	74.40%
10.1021/jacs.6b04077	25	-0.0240	+0.3308	+0.1451	60.58%	65.38%
10.1021/jacs.6b08236	47	-0.0074	+0.2589	+0.0690	70.22%	71.26%
10.1021/jacs.7b05460	35	-65.7868	-0.0227	+0.0912	74.91%	80.83%
10.1021/jacs.7b08207	40	-0.0886	+0.4265	+0.1877	51.88%	55.62%
10.1021/jacs.7b12628	28	n/a (near-zero variance)	-0.1157	+0.0217	85.71%	89.29%
10.1021/jacs.8b00178	85	-0.1216	+0.2504	+0.0679	60.65%	66.32%
10.1021/jacs.8b02865	35	-0.1865	+0.1142	+0.1394	52.08%	53.47%
10.1021/jacs.8b12095	29	n/a (near-zero variance)	-0.1083	+0.1050	90.00%	95.00%
10.1021/jacs.9b04332	40	-0.0529	+0.4193	+0.2120	61.25%	58.13%
10.1021/jacs.9b07514	38	-1.0372	-0.0881	+0.0247	48.03%	61.18%
10.1021/jacs.9b11938	37	-57.3029	-0.0392	+0.0420	42.11%	56.58%
10.1021/ol402877n	26	-1.5281	-0.0846	+0.0501	76.92%	80.77%
10.1038/ncomms14841	45	+0.0591	+0.4694	+0.2354	77.72%	77.17%
10.1038/ncomms15598	27	n/a (near-zero variance)	+0.0873	+0.0651	87.50%	90.18%
10.1038/s41467-018-04645-3	52	+0.3543	+0.6527	+0.4397	81.25%	53.37%
10.1038/s41467-018-	30	-0.3895	+0.3127	+0.1218	67.86%	65.47%

doi	n	r2	r	r2_aff	emo	emo_{base}
07462-w						
10.1038/s41467-019-13705-1	53	+0.2567	+0.5810	+0.3446	83.80%	83.33%
10.1038/s41467-020-17057-z	31	-0.4915	+0.1052	+0.0562	75.78%	75.89%
10.1038/s41467-020-19807-5	38	-0.4585	+0.0018	+0.1005	62.50%	67.76%
10.1038/s41467-021-22127-x	57	-0.2045	+0.4085	+0.1844	74.80%	73.92%
10.1038/s41467-022-28002-7	30	-0.6923	+0.1894	+0.2241	94.17%	94.17%
10.1038/s41557-019-0346-2	120	-0.2741	+0.0620	+0.0117	67.41%	69.70%
10.1038/s41929-020-00535-4	51	+0.0853	+0.4508	+0.2071	56.73%	58.17%
10.1038/s41929-020-0460-y	62	-0.0756	+0.4351	+0.2001	76.21%	71.77%
10.1039/C4CC03920C	30	-0.5606	-0.0987	+0.0631	70.42%	68.23%
10.1039/C6CC02742C	29	-0.4633	+0.1862	+0.1754	100.00%	100.00%
10.1039/C6CC08759K	28	-12.6882	+0.1362	+0.1029	58.04%	55.36%
10.1039/C8CC00493E	25	n/a (near-zero variance)	-0.0355	+0.0550	44.78%	58.26%
10.1039/C8CC03586E	30	-0.9827	-0.0282	+0.0557	87.50%	87.50%
10.1039/C9SC04073K	38	+0.2246	+0.6329	+0.4078	92.76%	60.53%
10.1039/D0CC00434K	26	-0.8172	+0.1102	+0.1353	75.96%	76.92%
10.1039/D0CC03201H	36	+0.0995	+0.4955	+0.2528	91.67%	91.67%
10.1039/D0CC04590J	32	n/a (near-zero variance)	-0.0055	+0.0653	80.47%	82.81%
10.1039/D0QO01380C	29	-19.3015	+0.5691	+0.3993	90.83%	84.17%
10.1039/D0SC04340K	25	n/a (near-zero variance)	+0.1696	+0.0891	72.12%	73.08%
10.1039/D0SC06162J	26	-0.5525	+0.1625	+0.1659	84.62%	85.58%
10.1039/D1CC01690C	27	-0.3768	+0.1508	+0.0615	57.14%	59.82%
10.1039/D1QO00212K	28	-0.3877	+0.2410	+0.0929	80.36%	80.36%
10.1039/D1QO00368B	27	-2.2893	-0.3385	+0.1626	89.29%	93.75%
10.1039/D1QO01310F	25	-1.6593	+0.1528	+0.1125	84.62%	92.31%
10.1039/D1QO01598B	28	-0.9230	+0.2701	+0.0890	57.14%	64.29%
10.1039/D1SC01096D	28	n/a (near-zero variance)	+0.1513	+0.1827	100.00%	100.00%
10.1039/D1SC01337H	36	-0.0378	+0.2020	+0.0733	65.28%	68.06%

doi	n	r2	r	r2_aff	emo	emo_{base}
10.1039/D1SC01891D	33	-1.1871	-0.2482	+0.0841	37.50%	60.29%
10.1039/D1SC06613G	64	+0.2398	+0.6116	+0.3868	69.14%	59.77%
10.1039/c7cc02159c	32	-0.1280	+0.3487	+0.1403	100.00%	100.00%
10.31635/ ccschem.019.20190064	57	-0.3544	+0.0674	+0.0514	78.88%	78.88%

⚠ In the raw `r2` column, **23** papers show $|R^2| > 100$ (the most extreme −2.48e18) — when the y variance of one half of a paper approaches zero the R^2 denominator approaches zero, which is an **unusable value**, marked `n/a` here; the analysis always uses the scale-free per-paper r.

Per-paper r: median **+0.1850**, quartiles +0.0424 / +0.1850 / +0.3617; share of papers with r>0 **80%**; row-weighted EMO **74.10%** vs the per-paper majority class **74.58%**.

S7-G Model-free between-condition comparison: all 66 condition pairs (`condition_pairwise.csv`)

`same_col`/`same_ph` indicate whether the pair shares the column/mobile phase; `r`/`rho` = Pearson/Spearman; `sign_agree` = sign-agreement rate.

pair	n	$same_{col}$	$same_{ph}$	r	rho	$sign_{agree}$
LA-1 / acidic ↔ LA-1 / basic	41	True	False	+0.4678	+0.7402	75.6%
LA-1 / acidic ↔ LA-2 / acidic	41	False	True	+0.0655	+0.2247	70.7%
LA-1 / acidic ↔ LA-2 / basic	40	False	False	+0.1519	+0.2814	52.5%
LA-1 / acidic ↔ LC-2 / acidic	37	False	True	+0.1195	+0.1961	73.0%
LA-1 / acidic ↔ LC-2 / basic	38	False	False	+0.0965	+0.0340	52.6%
LA-1 / acidic ↔ LC-3 / acidic	41	False	True	+0.1297	+0.1961	68.3%
LA-1 / acidic ↔ LC-3 / basic	40	False	False	+0.1124	+0.1176	50.0%
LA-1 / acidic ↔ LC-4 / acidic	41	False	True	+0.0347	+0.2428	75.6%
LA-1 / acidic ↔ LC-4 / basic	37	False	False	-0.2797	-0.1463	54.1%
LA-1 / acidic ↔ OD-RH / acidic	40	False	True	+0.0518	+0.0363	50.0%
LA-1 / acidic ↔ OD-	38	False	False	-0.2066	-0.2556	34.2%

pair	n	$same_{col}$	$same_{ph}$	r	rho	$sign_{agree}$
RH / basic						
LA-1 / basic ↔ LA-2 / acidic	42	False	False	+0.0174	+0.1527	52.4%
LA-1 / basic ↔ LA-2 / basic	41	False	True	+0.5421	+0.2874	56.1%
LA-1 / basic ↔ LC-2 / acidic	38	False	False	+0.0144	+0.0611	50.0%
LA-1 / basic ↔ LC-2 / basic	39	False	True	+0.0586	+0.0849	51.3%
LA-1 / basic ↔ LC-3 / acidic	42	False	False	+0.0843	+0.2298	54.8%
LA-1 / basic ↔ LC-3 / basic	41	False	True	+0.1273	+0.1464	48.8%
LA-1 / basic ↔ LC-4 / acidic	42	False	False	+0.0286	+0.0607	54.8%
LA-1 / basic ↔ LC-4 / basic	38	False	True	-0.0890	+0.0072	50.0%
LA-1 / basic ↔ OD-RH / acidic	41	False	False	-0.0271	-0.1127	29.3%
LA-1 / basic ↔ OD-RH / basic	38	False	True	+0.2809	-0.0981	42.1%
LA-2 / acidic ↔ LA-2 / basic	42	True	False	+0.0363	+0.3824	61.9%
LA-2 / acidic ↔ LC-2 / acidic	39	False	True	-0.1998	-0.0033	69.2%
LA-2 / acidic ↔ LC-2 / basic	39	False	False	-0.3309	-0.1405	48.7%
LA-2 / acidic ↔ LC-3 / acidic	43	False	True	-0.1916	+0.0726	74.4%
LA-2 / acidic ↔ LC-3 / basic	42	False	False	-0.2544	-0.1091	52.4%
LA-2 / acidic ↔ LC-4 / acidic	43	False	True	-0.3488	+0.0269	72.1%
LA-2 / acidic ↔ LC-4 / basic	39	False	False	-0.2230	+0.0144	51.3%
LA-2 / acidic ↔ OD-RH / acidic	42	False	True	-0.2512	-0.0403	69.0%
LA-2 / acidic ↔ OD-	39	False	False	-0.0568	+0.0167	35.9%

pair	n	$same_{col}$	$same_{ph}$	r	rho	$sign_{agree}$
RH / basic						
LA-2 / basic ↔ LC-2 / acidic	38	False	False	-0.0861	-0.0178	42.1%
LA-2 / basic ↔ LC-2 / basic	38	False	True	+0.3347	+0.1398	52.6%
LA-2 / basic ↔ LC-3 / acidic	42	False	False	-0.1080	-0.0142	50.0%
LA-2 / basic ↔ LC-3 / basic	42	False	True	-0.1723	-0.1454	54.8%
LA-2 / basic ↔ LC-4 / acidic	42	False	False	-0.0866	+0.0772	50.0%
LA-2 / basic ↔ LC-4 / basic	39	False	True	-0.2328	+0.0147	51.3%
LA-2 / basic ↔ OD-RH / acidic	41	False	False	+0.1594	+0.0887	48.8%
LA-2 / basic ↔ OD-RH / basic	38	False	True	+0.3387	+0.1641	52.6%
LC-2 / acidic ↔ LC-2 / basic	36	True	False	+0.7439	+0.6960	66.7%
LC-2 / acidic ↔ LC-3 / acidic	39	False	True	+0.5694	+0.5036	76.9%
LC-2 / acidic ↔ LC-3 / basic	38	False	False	+0.4593	+0.4421	55.3%
LC-2 / acidic ↔ LC-4 / acidic	39	False	True	+0.8995	+0.7670	92.3%
LC-2 / acidic ↔ LC-4 / basic	36	False	False	+0.5452	+0.4270	63.9%
LC-2 / acidic ↔ OD-RH / acidic	38	False	True	+0.1544	+0.4539	65.8%
LC-2 / acidic ↔ OD-RH / basic	35	False	False	+0.0838	+0.1770	37.1%
LC-2 / basic ↔ LC-3 / acidic	39	False	False	+0.3344	+0.3090	53.8%
LC-2 / basic ↔ LC-3 / basic	38	False	True	+0.2507	+0.3082	57.9%
LC-2 / basic ↔ LC-4 / acidic	39	False	False	+0.7972	+0.6180	66.7%
LC-2 / basic ↔ LC-4 /	37	False	True	+0.4276	+0.5910	73.0%

pair	n	$same_{col}$	$same_{ph}$	r	rho	$sign_{agree}$
basic						
LC-2 / basic ↔ OD-RH / acidic	38	False	False	+0.2869	+0.3353	44.7%
LC-2 / basic ↔ OD-RH / basic	35	False	True	+0.1088	+0.0575	40.0%
LC-3 / acidic ↔ LC-3 / basic	42	True	False	+0.6526	+0.5383	76.2%
LC-3 / acidic ↔ LC-4 / acidic	43	False	True	+0.3431	+0.2395	74.4%
LC-3 / acidic ↔ LC-4 / basic	39	False	False	+0.1950	+0.2264	56.4%
LC-3 / acidic ↔ OD-RH / acidic	42	False	True	+0.3142	+0.4540	69.0%
LC-3 / acidic ↔ OD-RH / basic	39	False	False	+0.2638	+0.2691	51.3%
LC-3 / basic ↔ LC-4 / acidic	42	False	False	+0.3321	+0.2914	59.5%
LC-3 / basic ↔ LC-4 / basic	39	False	True	+0.2044	+0.2198	53.8%
LC-3 / basic ↔ OD-RH / acidic	41	False	False	+0.2838	+0.3405	56.1%
LC-3 / basic ↔ OD-RH / basic	38	False	True	+0.1434	+0.0982	55.3%
LC-4 / acidic ↔ LC-4 / basic	39	True	False	+0.7663	+0.6340	74.4%
LC-4 / acidic ↔ OD-RH / acidic	42	False	True	+0.0621	+0.1432	61.9%
LC-4 / acidic ↔ OD-RH / basic	39	False	False	+0.1128	+0.1618	43.6%
LC-4 / basic ↔ OD-RH / acidic	38	False	False	-0.1331	-0.1116	36.8%
LC-4 / basic ↔ OD-RH / basic	35	False	True	+0.4360	+0.2001	42.9%
OD-RH / acidic ↔ OD-RH / basic	39	True	False	+0.1862	+0.2775	51.3%

Median r of the three classes: same column, different phase **+0.5602** (n=6 pairs) | different column, same phase **+0.1285** (n=30) | different column, different phase **+0.0840** (n=30). Per-column-pair values for "different column × same phase" (2 mobile phases per pair): `LC-2⇔LC-4` **+0.664** (sign agreement 82.6%) | `LC-2⇔LC-3` +0.410 | `LA-1⇔LA-2` +0.304 | `LC-3⇔LC-4`

+0.274 | LC-4⇔OD-RH +0.249 | LC-3⇔OD-RH +0.229 | LA-1⇔OD-RH +0.166 | LC-2⇔OD-RH +0.132 | LA-1⇔LC-3 +0.128 | LA-1⇔LC-2 +0.089 | LA-2⇔LC-2 +0.067 | LA-2⇔OD-RH +0.044 | LA-1⇔LC-4 −0.027 | LA-2⇔LC-3 −0.182 | LA-2⇔LC-4 **−0.291**.

S7-H Full external-validation table (`external_odrh_result.csv` + `external_all_columns.csv`)

train	n_{train}	n_{test}	r2	rmse	emo_{acc}	emo_{base}
all CMRT (across families)	10525	73	-0.4772	0.2575	43.84%	72.60%
CMRT `Cellulose-a` (selector-aligned)	2738	73	-0.8022	0.2844	47.95%	72.60%
all CMRT	10525	30	-0.3161	0.3568	36.67%	66.67%
CMRT `Cellulose-a`	2738	30	-0.6013	0.3935	46.67%	66.67%
CMRT `Cellulose-a` (acidic)	2738	38	-0.9580	0.2579	47.37%	71.05%
CMRT `Cellulose-a` (basic)	2738	35	-0.7085	0.3107	48.57%	74.29%

arm	column	family	n	r2	rmse	emo	emo_{base}	y_{sd}	$pred_{sd}$
OD-RH	OD-RH	cellulose	73	-0.4772	0.2575	43.84%	72.60%	0.2119	0.1272
LC-2	LC-2	cellulose	70	-0.3745	0.2532	34.29%	70.00%	0.2160	0.1199
LC-3	LC-3	cellulose	78	-1.4413	0.2006	34.62%	83.33%	0.1284	0.1260
LC-4	LC-4	cellulose	75	-0.4097	0.2554	38.67%	74.67%	0.2151	0.1270
LA-1	LA-1	amylose	75	-0.1490	0.3441	41.33%	74.67%	0.3210	0.1244
LA-2	LA-2	amylose	78	-0.3372	0.2452	42.31%	75.64%	0.2121	0.1260
OD-RH (aligned)	OD-RH [aligned]	cellulose	73	-0.8022	0.2844	47.95%	72.60%	0.2119	0.1620
LC-2 (aligned)	LC-2 [aligned]	cellulose	70	-0.4260	0.2579	51.43%	70.00%	0.2160	0.1665
LC-3 (aligned)	LC-3 [aligned]	cellulose	78	-1.9242	0.2195	55.13%	83.33%	0.1284	0.1679
LC-4 (aligned)	LC-4 [aligned]	cellulose	75	-0.5552	0.2683	52.00%	74.67%	0.2151	0.1693

Negative controls (target shuffled only, predictions kept in their original order): the OD-RH arm of `20` gives **−0.756 ± 0.157** (5 runs); the per-column shuffling of the 6 columns in `24` gives a mean of **−0.465 ± 0.284**, against a true mean of **−0.531**. Summarized by family: cellulose (4 columns) median R^2 **−0.443** (−1.441 ~ −0.375) | amylose (2 columns) **−0.243** (−0.337 ~ −0.149). Training-side convention: the selector-aligned arm = CMRT's `IB / OD / OD3 / ODH` (**2,738 rows**); the external side = DG25 `OD-RH` **88 rows minus 8 overlapping rows = 80 rows**, of which **73 rows** were successfully SMILES-parsed.

S7-I Zero-order correlation scan summarized by generation (full 196 columns in `zeroorder_scan.csv`)

generation	n_{signed}	n_{even}	med_{absr}_even	med_{absr}_signed	$signed_{gt}$_005	max_{absr}_signed
L1a-FF	48	48	0.0141	0.0189	3/48 = 6%	0.0616
L1-CF	61	28	0.0352	0.0169	1/61 = 2%	0.0791
L1b	7	4	0.0315	0.0107	0/7 = 0%	0.0290

Accurate wording: "**the signed columns carry no information beyond the even (unsigned) columns**"; **L1a-FF is the exception** (0.0189 > 0.0141, but both lie in the noise band), and this must not be written as "the signed columns of all three generations are worse".

S7-J L1a-FF per-column degeneracy and sign diagnostics (`l1a_per_column.csv`, 48 columns)

col	$zero_{frac}$	$sign_{eq}$_volmass
vol_{mass}	0.0000	1.0000
vol_{gast}	0.0002	0.4588
vol_{absq}	0.0002	0.9765
vol_{deg}	0.0000	0.9772
vol_{ring}	0.5309	0.9485
vol_{arom}	0.7596	0.8988
vol_{en}	0.0000	0.9806
vol_{cnt}	0.0000	0.9765
$voln_{mass}$	0.0000	0.9947
$voln_{gast}$	0.0002	0.4624
$voln_{absq}$	0.0002	0.9761
$voln_{deg}$	0.0000	0.9768
$voln_{ring}$	0.5357	0.9458
$voln_{arom}$	0.7709	0.9270
$voln_{en}$	0.0000	0.9797
$voln_{cnt}$	0.0000	0.9756
tp012_mass	0.0000	0.0261
tp012_gast	0.0002	0.5280
tp012_absq	0.0002	0.0371
tp012_deg	0.0000	0.0466
tp012_ring	0.6136	0.6275
tp012_arom	0.8006	0.7890
tp012_en	0.0000	0.0444

col	$zero_{frac}$	$sign_{eq}$_volmass
tp012_cnt	0.0000	0.0479
tp013_mass	0.0000	0.8541
tp013_gast	0.0002	0.4484
tp013_absq	0.0002	0.8803
tp013_deg	0.0000	0.8401
tp013_ring	0.9155	0.9792
tp013_arom	0.9620	0.9756
tp013_en	0.0000	0.8440
tp013_cnt	0.0000	0.8406
tp023_mass	0.0000	0.2114
tp023_gast	0.0002	0.5908
tp023_absq	0.0002	0.1899
tp023_deg	0.0000	0.2095
tp023_ring	0.9110	0.9392
tp023_arom	0.9480	0.9667
tp023_en	0.0000	0.2091
tp023_cnt	0.0000	0.2097
tp123_mass	0.0000	0.8054
tp123_gast	0.0002	0.5585
tp123_absq	0.0002	0.8006
tp123_deg	0.0000	0.8123
tp123_ring	0.9150	0.9910
tp123_arom	0.9462	0.9871
tp123_en	0.0000	0.8148
tp123_cnt	0.0000	0.8138

Columns with `zero_frac > 0.5` = **12** (`*_ring` / `*_arom`); of these, those with `zero_frac ≥ 0.9` = **6**. `vol_w = tp123_w − tp023_w + tp013_w − tp012_w` ⇒ the true rank of the 48 columns is ≤ **40** (8 columns are strictly linearly redundant).

S7-K. Elution-order accuracy of label-additive rules on the De Gauquier grid

Why it is needed: Section 3.5 of the main text states explicitly that "**variance shares and accuracies are not compared side by side**", so how good an additive description can be has to be measured directly in **accuracy language**. This table is evaluated only on **genuinely separated cells** ($\alpha \neq 1$, 215 cells) — because 56.5 % of the grid has $\alpha \equiv 1$ and $y \equiv 0$, and including those would turn the task into "can one predict 'not separated'", which is unrelated to elution order. Source scripts: `scripts/33b_emo_subset.py`, `scripts/33c_emo_order_ceiling.py` Outputs: `data/curated/review_round2_emo_subset.csv`, `review_round2_emo_order_ceiling.csv`

rule	n cells	EMO (%)	AUC (%)	notes
global majority class	215	**58.60**	50.00	constant predictor (positive class 58.60 %)
condition main effect only	215	**58.14**	56.14	majority direction from condition identity only
oracle two-way additive (molecule + condition dummies, no interaction)	215	**72.09**	76.13	**in-grid upper bound for the label-additive class**
leave-one-molecule, condition known	215	**76.28**	83.97	honest additive baseline (the molecular effect of an unseen molecule is set to the global mean)
per-molecule oracle majority	215	**77.67**	85.35	upper-bound reference: the molecule's majority direction is known
per-molecule majority (weighted mean)	215	77.63	—	computed per molecule first, then averaged

Reference: on the same 494-cell grid, cells with `α ≡ 1` account for **56.48 %** (`review_round2_emo_subset.csv`).

How to read it:

1. "**oracle main-effects R² = 23.05 %**" is the variance share of the **labels**, **not an accuracy ceiling**: on the same grid the additive rules reach an EMO of **72.1 %** (oracle) and **76.3 %** (leave-one-molecule) on separated cells.
2. The "structural explanation" of the paper is therefore **not** "no additive model can do it", but "**h is not identifiable in this data structure** (each (molecule, condition) cell has exactly 1 observation), and cross-study condition metadata and replicate measurements are missing".
3. ⚠ Single dataset (DG25, 43 molecules, 1 laboratory) ⇒ **it cannot be extrapolated as a bound for CMRT**.

S7-L. PE25 external validation (not independent of DG25; reported for completeness only)

Why it is listed separately: DG25 and PE25 **share 42/43 compounds** (Section 2.4), so they are **not two independent external sets**; the main text therefore uses **DG25 only** as the external validation, and PE25 is reported here for completeness. It **must not** be taken as independent corroboration. Source script: `scripts/10_external_validate.py` Output: `data/curated/external_pe25_result.csv`

subset	n pairs	R²	RMSE	EMO (%)	majority baseline (%)	y SD
all external pairs (incl. non-separated)	84	−0.1687	0.2802	47.6	69.0	0.2592
after removing CMRT-	**78**	**−0.1732**	0.2900	**48.7**	**69.2**	0.2678

subset	n pairs	R^2	RMSE	EMO (%)	majority baseline (%)	y SD
overlapping pairs separated only ($\alpha \neq 1$)	42	−0.0207	0.3631	52.4	57.1	0.3594

Convention: molecular-side features (25 dimensions, no condition/CSP); `HistGradientBoostingRegressor(max_iter=250, lr=0.06, min_samples_leaf=20, random_state=42)`; the 4 analytes overlapping CMRT (Ibuprofen / mandelic acid / modafinil / rolipram) were removed. **The conclusion agrees with DG25: cross-study R^2 is entirely negative and EMO is below the majority-class baseline.**

S8. The two negative controls: details and their own lessons

S8.1 Negative control 1: shuffled-label control

What it is meant to answer: whether a metric that "looks good" is signal or a pipeline artefact.

Correct procedure: **shuffle only the target `y`, keep the predictions `p` in their original order**, then run the **exactly identical pipeline**.

🔴 **A pitfall we fell into ourselves (`scripts/20`, first version)**: the negative control was written to permute `p[mask]` and `yte[mask]` together ⇒ **the pairing was not destroyed at all**, and 5 shuffles gave an R^2 **identical** to the original (-0.802 ± 0.000): it **looked "very clean" while in fact testing nothing**. Shuffling the target only gives $\mathbf{-0.756 \pm 0.157}$.

Lesson: **a negative control must be shown to actually destroy the signal**. Criterion: if the numbers are **identical** before and after shuffling, that is a **red flag**, not clean evidence; the negative-control result should be **indistinguishable from chance**, or at least fall within the interval consistent with the random expectation.

List of negative controls already implemented in this study:

scenario	negative-control result	source
S2c EMO (3 shuffles)	**50.17% ± 0.22%** (52.85% is +2.68 pp above it); row-level null, superseded by the within-paper null of S11.5	`s2c_emo.csv` / `negative_controls.csv`
full-table modelling (shuffled labels)	$\lvert R^2 \rvert \le 0.064$, EMO ≈ 50.2%	Section 2.7
predictability of the literature offset (shuffled paper means)	$R^2 = \mathbf{-0.0041}$ (clean, no leakage)	—
condition-aligned external validation (target shuffled only)	$\mathbf{-0.756 \pm 0.157}$	`negative_controls.csv`
cross-selector 6 columns (per-column shuffling)	mean $\mathbf{-0.465 \pm 0.284}$ vs the true mean $\mathbf{-0.531}$ ⇒ **the true value is no better than random**	`negative_controls.csv`
three-generation descriptor ablation	every arm is paired with a shuffled control (negative $R^2 \approx -0.05 \sim -0.07$, EMO ≈ 50%)	`06b/06d/14`

S8.2 Negative control 2: self-test of the numeric gate (a gate must verify that it fails)

Background: `scripts/26_paper_number_gate.py` asserts that "the numbers cited in the main text/figures = the numbers in the CSVs on disk". A gate that **never fails** is equivalent to no gate.

Procedure: (i) **deliberately corrupt one expected value** and confirm that the script **exits non-zero** (this was done historically, but **no reproducible proof was left in the repository**); (ii) **the discriminating power of the criterion itself must be demonstrated** — see below.

Current state: **95 assertions, 0 failures** (`python scripts/26_paper_number_gate.py`; hardened on 2026-09-18 from 45 assertions, with new fold-count/convention/dirty-value assertions).

🔴 **Self-test of the criterion (the most important lesson of this section)**: `--trace` matches every **decimal** in the main text against a pool formed from "every numeric cell of 50 CSVs plus ×100/÷100"

(about **3.9 million** numbers) with a tolerance of ±5e-4. In practice this criterion has **almost no discriminating power**:

null control (random numbers fed into the same criterion)	misjudged as "traceable"
uniform [0,1), three decimals	**100.0%**
uniform [0,100), two decimals	**100.0%**
log-uniform 1e-3…1e3	93.2%

⇒ **"0 untraceable" ≠ "every number has a provenance"**; it only shows that the pool is too dense. The same holds at the integer level (all 51 integer tokens lie among the 140,537 pool values, and even a value known to be wrong, such as `523`, matches some cell that happens to equal 523). ⇒ **The real safeguard is the 95 item-by-item assertions** (fold counts, conventions, dirty values); `--trace` can only serve as a **weak screen**; the script now **reports** its own discriminating power and fails under `--strict`.

Lesson: this study has repeatedly encountered **silent failures**, in five related forms: (i) `except: fallback value` silently turned a chemical-identity API failure into a random value (`06`, first version: agreement 92.3% → 55.6%); (ii) `CalcMurckoScaffold` did not exist → `try/except` collapsed all scaffolds to the same value; (iii) the negative control was a no-op (S8.1); (iv) in `18`, `if col not in p.columns: continue` **silently skipped every control split** and produced an empty table; (v) the global whitespace normalization `re.sub(r"\s+"," ",text)` in `29` squeezed the whole manuscript into one line, and the script still reported "0 residues" ⇒ fixed by allowing newlines inside the regex plus **row/section-count assertions before writing**. ⇒ **Rule: a gate or a control must be demonstrably able to fail; any `except: skip` must raise an error or at least log.**

S8.3 Companion tool: power curve — turning "zero" into a bounded negative

A null result is interpretable only when accompanied by a power statement. Procedure: **artificially implant** a composition-driven offset with a known variance share and see how much of it **the same pipeline** recovers:

implanted compositional explainable share	mean ± SD recovered over 5 noise draws	recovery rate
0.05	**+0.0195 ± 0.0168**	39%
0.10	**+0.0737 ± 0.0097**	74%
0.20	**+0.1558 ± 0.0256**	78%
0.39	**+0.3637 ± 0.0164**	93%

The observed value is $\mathbf{R^2 = -0.0158}$. Using the conservative criterion adopted in the main text (recovery mean − t_4 × draw-to-draw SD, $t_4 = 2.132$ for a one-sided 95 % bound on five draws), the lower bound is **−1.63 %** at a planted share of 5 % (below the observed value, so the 5 % level is **not** excluded) and **+5.30 %** at 10 % (above it, so shares **≥ 10 % are excluded**). A bound on the standard error of the mean (2.2 times narrower) would instead exclude the 5 % level, but that is not the criterion used in this paper. **Boundary (must be reported alongside)**: recovery is below 100% (the implanted signal uses only 2 features and there are only 530 papers) ⇒ the curve gives a **lower bound on sensitivity**, not an exact power

calculation. **Wiring check**: when the target is a column already contained in the paper means (`cond_ipr`), R^2 = **+1.0000** ⇒ the pipeline is connected.

S8.4 Permutation nulls on the S2b protocol

Why it is needed: the negative control in `scripts/18` permutes only **3 times** and shuffles at the **row level** — it destroys each paper's class proportions, which is inconsistent with the paper-aggregated S2c metric, and it cannot supply a null hypothesis for "is +2.18 pp significant". Source script: `scripts/37_study_blocked_null.py` | output: `data/curated/s2c_null_studyblocked.csv`

Protocol: S2b (leave-a-batch-of-papers, 10 folds) — a **cheap proxy** for leaving out whole papers (7 s per training run, versus about 3 h for the 530 folds of S2c). The label blocks of each paper are permuted under two schemes, and **the classifier is retrained after every permutation** (⚠ using a "constant majority-class predictor" yields a **degenerate null distribution** with SD = 0; the first version of this script fell into that trap, and this is documented in the docstring). Observed value: **S2b EMO = 53.938 %**.

scheme	n_{perm}	null mean (%)	null SD (pp)	95 % range (%)	z	empirical p(≥obs)
within-study (per-paper class proportions preserved; label↔feature pairing destroyed)	200	**51.79**	0.63	50.71–52.88	**+3.41**	< 0.005
study-block (whole-paper label blocks permuted between papers)	200	**50.54**	0.55	49.51–51.67	**+6.15**	< 0.005

How to read it:

1. The **within-study** permutation is a conservative null hypothesis with the correct convention (the model may legitimately exploit each paper's class proportions); on S2b, z ≈ 3.4 ⇒ the residual signal is detectable, but its size is only **+2.15 pp** (53.94 − 51.79).
2. S2b is only a cheap proxy for holding out whole papers. The null computed directly on the main protocol (S2c, 530 folds, 30 within-paper permutations with full retraining) is reported in **S11.5**: null 52.33 % ± 0.58 pp against the observed 52.85 % (z ≈ 0.89, empirical p ≈ 0.13), so the +2.15 pp seen on S2b does not survive on S2c. The main text therefore describes the S2c accuracy as indistinguishable from the within-paper null, and uses the S2b result only as the cheaper cross-check.
3. **Terminology**: the S2c null permutes labels **within each study** (the *within-study* scheme in the table above); the separate **study-block** scheme permutes whole per-paper label blocks between studies and is not used for S2c (see S11.5).

S9. Reproducibility gate sequence

Windows + pwsh; the console defaults to GBK, so **`PYTHONIOENCODING=utf-8` must be set before every python invocation**; install packages only with `python -m pip`.

```
cd "$(git rev-parse --show-toplevel)"    # the repository root
$env:PYTHONIOENCODING = "utf-8"

# ── (i) Data gate: ensures the cleaning specification has not been broken (requires the
upstream CMRT raw download)
python scripts/01_curate_cmrt.py
python scripts/_qa_01_curate.py          # expected: all 67 checks pass

# ── (ii) Paper-number gate: ensures the numbers cited in the paper match the CSVs on disk
python scripts/26_paper_number_gate.py   # expected: 95 pass, 0 fail
python scripts/26_paper_number_gate.py --trace            # weak screen; reports its own
discriminating power (random three-decimal numbers are 100% misjudged)
python scripts/26_paper_number_gate.py --trace --strict    # expected to fail by design: the
criterion self-test reports 100 % degenerate discriminating power

# ── (iii) Regeneration of the main results (each script writes its own note + CSV)
python scripts/17_reproducibility_ceiling.py     # 24 s    between-paper variance / power
curve / offset predictability
python scripts/19_paper_learning_curve.py        # 34 s    within-laboratory vs change of
laboratory
python scripts/18_s2c_emo.py                     # 337 s   S2c EMO (classification arm)
python scripts/21_condition_reproducibility.py   # fast    model-free between-condition
comparison
python scripts/22_interaction_ceiling.py         # ~1 min  additive ceiling + bootstrap
python scripts/20_external_odrh.py               # fast    selector-aligned external
validation
python scripts/24_external_all_columns.py        # fast    cross-selector 6 columns
python scripts/25_t0_sensitivity.py              # ~19 min dead-time sensitivity
python scripts/15_leave_one_doi.py               # ~13 min S2c fold indices + R²_aff ≡ η²
# other descriptor arms: scripts 06b / 06d / 14 / 16 (6–10 min each; need the descriptor
tables from 06/06c/12)

# ── (iv) Figures and manuscript
python scripts/38_make_figures_v3.py     # the seven publication figures (reads the CSVs on
disk only; recomputes no number)
# The manuscript sources and the built PDF/DOCX are shipped under manuscript/ and preprint/;
# rebuilding them additionally requires python-docx and a LibreOffice kit, and is not needed
# to reproduce any number or figure in the paper.
```

Order dependence: `01` → `17/19/21/22/20/24/25/15/18` → `38` → `26`. `06/06c/12` are the upstream of the descriptor tables (about 12 min / 12 min / fast each) and only need to be run once before the descriptor ablation. **Environment versions**: Python 3.14.3 / RDKit 2026.03.6 / scikit-learn 1.9.0 / NumPy 2.4.6 / pandas 3.0.3 / SciPy 1.18.0; fixed seed `SEED = 42`, geometric seed `0xC1A1`.

Licences and access for external dependencies: CMRT (Zenodo `10.5281/zenodo.7623903`, CC BY 4.0, **derivative works must be attributed**) | DG25 (`10.5281/zenodo.16636955`) | PE25 (`10.5281/zenodo.13682760`, record year 2024) | CMCDS (Figshare `10.6084/m9.figshare.28805909`, CC BY 4.0; the local egress IP was once blocked with a 403 by its WAF, so another network is required; file MD5 `181c1015f2befeeedcf15da5d8494650`).

S10. The ten limitations and their recommended interpretation (full version)

#	limitation	recommended wording / why it matters	source
1	**The same-molecule matched subset accounts for only 2.6–3.8% of the full table**, and systems that	Can only be written as "on the **strict same-molecule matched subset**, the cross-paper SD is of the same	`reproducibility_components.csv`

#	limitation	recommended wording / why it matters	source
	"can appear in ≥2 papers" are themselves a minority ⇒ **selection bias; not extrapolable**	order as the total SD of the data ⇒ the R^2 ceiling **for such rows** ≈ 0.32–0.37"; **must not** be written as "all of the 38.5% is a laboratory effect"	
2	**CMRT has no replicate measurements**: the median number of rows per molecule is **1**; the median of `kS_rsd`/`kR_rsd` is 0 (**96.2%/96.4% are exactly 0**, because the corresponding side has `k_n = 1`) ⇒ **"hard to predict" cannot be distinguished from "the target variable itself is noisy"**	State explicitly that "measurement noise cannot be estimated from within the dataset"; ⚠ **do not write "RSD is always 0"** (measured: 3.8% non-zero, n=396, median 0.0546, maximum 0.947)	main-table columns `kS_rsd`/`kR_rsd`
3	**The interaction-term decomposition comes from a single dataset** (DG25: one laboratory, 43 molecules) ⇒ **it is a proof-of-principle demonstration, not a universal constant**; moreover, **exactly 1 observation per cell** ⇒ **interaction and noise are inseparable**; the grid is **95.7% complete and unbalanced** (494/516, n = 494)	**23.05%** (bootstrap distribution: 95 % percentile intervals [17.5 %, 32.9 %] resampling molecules and [18.2 %, 40.8 %] resampling molecules and conditions; distributional summaries, not centered confidence intervals) must be written as an **upper bound**; it must not be read as a bound for CMRT (CMRT molecules are more heterogeneous, so the molecular main effect will be **smaller**); three sensitivity analyses give 22.29% / 22.33% / 20.18% ⇒ the convention spans 20.2–23.1% and the direction of the conclusion is unchanged	`interaction_ceiling.csv`
4	The "composition features" of evidence B **do not include the stationary-phase identity** ⇒ the conclusion is limited to "the offset is unpredictable **under molecule + mobile-phase descriptors**"	Add that the stationary phase does matter (Sections 3.4 and 3.6), but its role is **re-ordering**, not translation	`paper_offset_predictability.csv`
5	The external validation has only **73 rows**, and the **mobile-phase mode is entirely different** (normal phase in CMRT vs water/ACN in DG25) ⇒ this is a **compound** shift and **cannot be attributed to the laboratory alone**; "unseparated" samples account for **49–72% per column (58% overall; DG25 test subset 449 rows)** (CMRT: only 0.16%)	Must state "a compound shift across conditions within the same selector, the two being inseparable"	`external_dg25_all.csv`, `external_odrh_result.csv`, `external_all_columns.csv`
6	**The within setting has only ~15 training rows** ⇒	State explicitly that "it will rise as more data become	`within_paper_split.csv`, `learning_curve_papers.c`

#	limitation	recommended wording / why it matters	source
	+0.185 is "the ability at small sample size", **not an upper bound**	available within a paper"; by contrast, the flatness of the across curve **does not depend** on the training size of any single fold	`sv`
7	**t_0 is an estimate** (the dataset contains no experimental dead time; φ = 0.62; all columns 4.6 mm × 250 mm)	✅ A scan over φ ∈ {0.55, 0.58, 0.62, 0.66, 0.70} has been added: `y` against the reference gives **r ≥ 0.99444**, EMO sign flips **≤ 0.01%**, and the S2c R^2 is **negative throughout** (−0.0465 ~ −0.0688) ⇒ the conclusion is unchanged; ⚠ however **R^2 moves by about 0.02 in magnitude** (comparable to the effect size of interest here) ⇒ this **limits the absolute reading of R^2**, and measuring the dead time experimentally in future work is recommended	`t0_sensitivity.csv`
8	**The "per-paper majority-class oracle baseline of 76.55%" in S2c is close to the 76–96% often reported in the literature** ⇒ it **can only serve as "the convention must be clarified" and must not be written as a causal accusation**	State plainly that "different datasets/splits/models make this a coincidence at most, which we cannot explain", and close the conclusion at the level of **reporting practice** (EMO must always be reported together with the per-paper majority-class baseline)	`s2c_emo.csv`
9	**149 molecules (1.5%) admit no descriptor**, almost all of them **sulfur stereocentres (sulfoxides, Z=16) ⇒ a design-driven exclusion**	The scope and reason of the exclusion must be declared (`l1a_failures.csv` / `l1cf_failures.csv`)	descriptor-failure table
10	**The values declared in the paper are inconsistent with the local measurements**: the CMRT paper declares **25,847 molecules / 11,720 enantiomer pairs**, whereas locally we obtain **25,866 records / 10,525 pairs**	**State both numbers side by side with their sources**, and explain that the difference comes from the cleaning specification (especially the removal of 597 diastereomer pairs)	—

Additional disclosures (S10 supplement): (i) `doi_raw` is not faithful in the main table (it equals `doi` row by row); the original string survives only in `cmrt_measurements.csv`; (ii) **1** record has the literal DOI `"s"` (unrepairable); (iii) the 20 excited states of CMCDS **cover the far ultraviolet only** (median λ 187 nm; only 55.4% of molecules have a transition with λ > 250 nm) ⇒ not used for spectral prediction; (iv) the coverage of L1b differs from that of L1-CF/L1a (98.2% vs 98.5%), so the sample set changes with the generation in cross-generation comparisons.

S11. Additional analyses

This section contains the five additional analyses (H1, H3, S3′, H7 and H2); all of them read existing data only and none re-ran the main chain; scripts `78`–`80`, with every number generated from the CSVs on disk. Aim: to turn three "weak" conclusions (attribution of the within-paper signal, significance of the main protocol, attribution of the external failure) into defensible strong ones.

S11.1 Attribution of the within-paper signal: molecular or condition? (H1)

Question: the within-paper 50/50 split groups by molecule only, so training and test **share conditions** (across the 170 papers, a median of 10 (column, ipr, flow-rate) combinations, 4 columns and 5 ipr levels) ⇒ the model might "cheat" with the condition features; and t_0 enters both the features and the target.

Approach: the **same batch of 1360 splits** as the main arm of Section 3.3 (170 papers with n≥25 × 8 repeats of 50/50, Ridge(alpha=10) + LogReg(C=0.1)), changing only the feature family.

feature family	median per-paper r	share of papers with r>0	EMO (row-weighted)	EMO baseline
molecules only (25 dims)	**+0.1691**	79%	73.96%	74.58%
conditions only (ipr/speed/t0)	+0.1104	61%	73.32%	74.58%
molecules + conditions (= L0, anchor)	**+0.1850**	**80%**	74.10%	74.58%
molecules + conditions (without t0)	+0.1856	79%	74.21%	74.58%

- Anchor check: molecules + conditions = **+0.1850**, digit-for-digit identical to the +0.185 of `within_paper_split.csv`.
- Paired tests: adding the condition features gives **Δr = −0.0001 (Wilcoxon p = 0.41, n=170)**; molecules vs conditions **p = 0.0092**; removing t_0 gives **Δr = +0.0006**.
- **Conclusion**: the within-paper signal is **carried by the molecular description (91% of its magnitude)**; it is not a product of shared conditions or of t_0 leakage; the attribution of C3 holds.

S11.2 Confidence intervals for the protocol EMO (H3)

Approach: recompute the OOF for the four protocols with the same classifier/features/folds as `scripts/18` (all reproduction assertions pass: 66.1283/71.7910/53.9382/52.8456%), then run a 2,000-draw bootstrap **clustered by paper (DOI)**.

quantity	point estimate	95% CI	SD
EMO random (S1)	66.13%	[64.55, 67.82]	±0.81 pp
EMO grouped (S2)	71.79%	[70.18, 73.52]	±0.86 pp
EMO leave-a-batch-of-papers (S2b)	53.94%	[51.73, 56.18]	±1.12 pp
EMO true leave-one-paper-out (S2c)	**52.85%**	**[50.63, 55.15]**	±1.15 pp
S2c − S2b	−1.09 pp	[−2.44, +0.21]	indistinguishable
S2c − S2	**−18.95 pp**	**[−21.31, −16.59]**	significant
S2c − S1	**−13.28 pp**	**[−15.58, −10.98]**	significant
S2c gain over the global	**+2.18 pp**	**[−0.04, +4.48]**	one-sided P=0.028

quantity	point estimate	95% CI	SD
majority class (50.67%)			

- **Conclusion**: the protocol effect (−19 pp) is very stable under paper-level resampling; the CI of "+2.18 pp above the majority class" **touches 0 at its lower edge**, i.e. it lies at the **edge of resolution** under the 95% convention, and must be read side by side with the permutation nulls: on S2b the residual is +2.15 pp ($z \approx 3.4$), whereas on the main protocol S2c $z \approx 0.89$ and the residual is not distinguishable from chance (S11.5). It must not be reported as "detectable" alone.

S11.3 Leave-one-selector-family extrapolation (S3′)

Six directions (amylose ↔ cellulose, three subsets a/b/f), with R^2 negative throughout:

direction	vendor labels	self-built CSP full set	substituent structure only
$S3p_a$ amylose→cellulose	−0.160	−0.156	−0.155
$S3p_a$ cellulose→amylose	−0.180	−0.171	−0.169
$S3p_b$ amylose→cellulose	−0.173	−0.173	−0.173
$S3p_b$ cellulose→amylose	−0.104	−0.104	**−0.019**
$S3p_f$ amylose→cellulose	−0.660	−0.660	**−0.156**
$S3p_f$ cellulose→amylose	−0.297	−0.297	−0.297

- **Conclusion**: **all six directions are negative**; even under the best CSP-descriptor convention R^2 remains **−0.019 to −0.297**; four directions improve by ≤0.010. The extrapolation failure **is not a matter of selector description**.

S11.4 Leave-one-paper-out within each column (H7)

arm	EMO (row-weighted)	baseline	notes
arm A: within-column random (literature style)	74.61%	55.83%	contains paper leakage; control only
arm B: within-column leave-one-paper-out (strict)	**52.73%**	83.76% (per-column oracle majority class)	collapse persists when papers change within the same column
B − A (per column)	median **−20.5 pp**	—	**24/25 columns negative, 1 column at 0 (IAH, n=19), 0 columns positive**

- **Conclusion**: **within a single column**, changing only the papers still makes the model collapse ⇒ the external failure **is not an artefact of mixing "column × mobile phase"**.

S11.5 Permutation null for S2c (530 folds) (H2)

Convention: S2c with 530 folds; each permutation **shuffles the EMO within each paper** (preserving each paper's class proportions — what the paper calls the conservative and relevant null), and **530 HGB classifiers are retrained every time** (same configuration as `scripts/18`). B=30 permutations, 14 processes in parallel, written to disk incrementally after each permutation.

quantity	value
null EMO mean ± SD	**52.326% ± 0.583 pp**
null 95% interval	[51.357, 53.363]
observed S2c EMO	**52.846%**

quantity	value
z	**+0.89**
empirical p (one-sided, B=30)	**0.133** (resolution 1/31 ≈ 0.032)

- **Conclusion**: **on the main protocol (S2c), the observed 52.85% is indistinguishable from the within-paper permutation null (52.33% ± 0.58) (p = 0.13) ⇒ no detectable residual skill** on the main protocol.
- The +2.15 pp residual seen on the S2b proxy (z ≈ 3.4) **does not reproduce on the main protocol**: S2c is only +0.52 pp above its null (z ≈ 0.89).
- The reason the null sits at 52.3% (rather than the global majority class of 50.67%): the within-paper permutation preserves each paper's class proportions, so the model can still learn the "study-level class balance" from the condition features. ⇒ **the +2.18 pp gain over the global majority class comes mainly from the condition structure, not from molecular-side skill**; this is consistent with across-paper r = +0.0009 and with the H3 CI [−0.04, +4.48].
- **Consequence**: the abstract and Section 5 therefore report the S2c accuracy as indistinguishable from the within-paper permutation null on the main protocol; this is a **stronger** negative conclusion, not a weaker one.
- **On block permutation**: `--scheme block` was tried and then abandoned for S2c — the number of rows differs between papers, so "whole-block permutation" has mismatched block sizes and a non-unique convention; the block-scheme value reported for S2b (50.54% ± 0.55) comes from `scripts/37` and was not re-verified on S2c. **The conclusions of this section rest on the within-paper permutation only** (the kind the paper itself calls conservative and relevant).

S11.6 Index of new artefacts

type	file
scripts	`scripts/78_within_paper_attribution.py`, `scripts/79_protocol_emo_ci.py`, `scripts/80_s2c_null_parallel.py`
data	`data/curated/within_paper_attribution.csv`, `data/curated/protocol_emo_bootstrap.csv`, `data/curated/upgrade_oof_preds.csv`, `data/curated/s2c_null_530fold.csv`